

GammaLoop

Differential cross-sections with Local Unitarity

1 Overview

GammaLoop computes differential collider cross-sections with the Local Unitarity method. It is an application and research codebase rather than a thin numerical library: process generation, model import, integrand construction, integration, observables, persistence, and diagnostics meet in one stateful workflow.

Primary interface

The command-line interface is the primary user interface. A run card creates a state, executes commands, and leaves a reusable working directory. The Rust and Python APIs load the same persisted state and expose selected structured operations; they do not define a second, independent execution model.

1.1 Choose a task

- To create and resume a known state, follow the [first-state tutorial](#).
- To adapt a process specification or generation filters, use the [process-generation guide](#) with the [generated generate reference](#).
- To inspect concrete cut events, selectors, and histogram snapshots without a full integration, follow the [events and observables guide](#).
- To automate a loaded state or diagnose a long run, choose the [Rust/Python interface guide](#) or [logging and diagnostics guide](#).

1.2 A stateful run

A typical source checkout is built and exercised from the repository root:

```
just build-cli-release
./gammaLoop ./examples/cli/gg_hhh/1L/gg_hhh_1L.toml
```

The run card imports a model, generates the requested process, executes its command blocks, and records the resulting state. Resume that directory explicitly for subsequent work:

```
./gammaLoop -s ./examples/cli/gg_hhh/1L/state \  
run integrate_physical -c "quit -o"
```

The persisted run.toml records how to replay the run. The settings files describe global and default runtime configuration, while processes/ contains process-specific state. Ordinary runs create gammaLoop_state/ unless a different state path is supplied.

1.3 Installation and external tools

GammaLoop is currently developed primarily from source. The supported development path is the repository's Nix shell, or a local Rust toolchain together with just, a recent GNU toolchain, and Python 3.11 or newer when building bindings. FORM 4.2.1 or newer is needed for analytical integration of integrated UV counterterms. UFO model import also needs the Python ufo-model-loader package.

```
nix develop
just build-cli
just build-api
```

Diagram rendering is a separate concern and uses Clinnet and Typst. Building the CLI does not imply that these drawing tools are installed. Use the [Clinnet DOT-rendering guide](#) for their independent versions, templates, and cache boundary.

1.4 Related crates

How the crates fit together

Linnet provides the half-edge graph model and graph algorithms. Spenso provides typed tensors, tensor structures, and network execution. Idenso handles symbolic tensor identities and algebraic simplification. Vakint matches and evaluates vacuum-integral topologies. GammaLoop combines these components into a collider-calculation workflow.

For a deeper view of how the command layer, state, integrands, and external components fit together, see the [GammaLoop implementation architecture](#).

1.5 Where to begin

- Use built-in `--help` for the CLI options supported by your installed version.
- Start with the included `gg_hhh` run card for a complete state lifecycle.
- Use the Rust or Python API only when a program needs structured access to loaded state or sample-evaluation results.
- Check prerequisites before starting an expensive integration; some workflows require a Symbolica license or external tools such as FORM.

2 Tutorial

This tutorial builds the GammaLoop command-line interface and uses the included one-loop `g g -> h h h` run card to create a state that can be inspected and resumed. Compilation and process generation can take longer than a `--help` smoke test.

2.1 Prerequisites

Work from a GammaLoop source checkout. The supported development environment is the repository Nix shell; without Nix you need Rust 1.85 or newer, `just`, a recent GNU toolchain, and the dependencies described in the root README. The bundled Standard Model used below does not need the UFO loader. A Symbolica license may be required by your build and use case.

```
nix develop
just build-cli
./gammaloop --help
```

The last command should print the one-shot CLI options and command tree. The `./gammaloop` wrapper selects the binary built under `target/dev-optim/`.

Verification scope and cost

The docs harness checks these shell blocks for syntax; it does not build or execute GammaLoop. Verify a clean checkout manually in `nix develop`. A cold CLI build and process generation are the high-resource path and can take tens of minutes, while reopening a valid state is normally much cheaper. The invariant for this tutorial is a resumable state with the files listed below, not a completed Monte Carlo integration.

2.2 Generate an example process

The run card declares a state folder, settings, and named command blocks. Run its generate block and persist the result on exit:

```
./gammaloop --clean-state \  
./examples/cli/gg_hhh/1L/gg_hhh_1L.toml \  
run generate -c "quit -o"
```

--clean-state removes the resolved state first

Use it only for this first, reproducible run. Omit it when resuming work you want to keep. Its generated argument reference records the exact valueless-flag semantics. The example card resolves its state to examples/cli/gg_hhh/1L/state and requests ten worker cores; make a private copy of the card before changing either value.

The block imports sm-default.json, generates the selected one-loop pentagon contribution, builds its integrand, saves DOT data, and writes the state. A successful run exits without an error and leaves at least run.toml, global_settings.toml, default_runtime_settings.toml, and processes/ in the state folder. run.toml records the commands and settings needed to replay the run; it is not merely a log file.

2.3 Resume the saved state

Load the saved state and ask the active session to display its processes:

```
./gammaloop -s ./examples/cli/gg_hhh/1L/state \  
run -c "display processes; quit -o"
```

Success means the display includes the generated gg_hhh process and the command exits while keeping the same state. You can now run the card's integrate_euclidean or integrate_physical block, but those are deliberately beyond the first-success path because they request a substantial Monte Carlo integration.

2.4 Inspect before expensive work

Reopen the state read-only and inspect both the generated structure and the effective integrator settings before launching either integration block:

```
./gammaloop --read-only-state \  
-s ./examples/cli/gg_hhh/1L/state \  
run -c "display integrand -p gg_hhh -i 1L; display settings process -p gg_hhh -i 1L  
integrator; quit"
```

display integrand reports the generation backend, compiled record size, graph and graph-group counts, and detailed orientation, loop-momentum-basis, cut, and threshold tables. The settings command resolves the values attached to this generated integrand, which may differ from the current defaults. Check these records together: an unexpected graph count points back to process generation, while an unexpected sample budget, target accuracy, or seed belongs to runtime settings.

Read-only inspection is not persistence

--read-only-state prevents commands from writing inside the active state directory and disables file logging there. It does not make the in-memory session immutable. This command intentionally exits without -o; change settings in a separate writable session or in the run card that reconstructs the state.

The exact fields and selectors are kept current in the generated integrand display and process-settings references.

2.5 Troubleshooting and next steps

- If the wrapper cannot find a binary, rerun `just build-cli` and confirm that `target/dev-optim/gammloop` exists.
- If compilation or process generation exhausts local resources, copy the card and reduce the values under `cli_settings.global.n_cores`; do not edit a state midway through generation.
- If startup reports a state fingerprint or settings mismatch, replay the card with a new state folder instead of transplanting only its `processes/` directory.
- Use `./gammloop help generate` to inspect the flags supported by your installed version, then continue with the [process-generation guide](#) and the [generate-command reference](#).

Example run card

See the complete `gg_hhh` run card before adapting the process, state location, or resource settings for your own calculation.

3 Process generation and state workflows

GammaLoop describes a calculation with a TOML run card and stores its reusable state in a directory. Start the command-line interface through the repository wrapper (`./gammloop`) or the Cargo-built executable, then generate processes, integrands, and integration jobs within that state.

3.1 Choose a generation mode

The interactive command tree separates cross sections, amplitudes, and operations on processes already stored in the active state:

```
generate xs INITIAL > FINAL ...
generate amp INITIAL > FINAL ...
generate existing ...
generate help xs
```

`xs` constructs forward-scattering graphs and filters compatible Cutkosky cuts. `amp` constructs ordinary amplitudes. Their final-state lists therefore have different meanings even when the printed particle syntax is identical. Use `generate help <mode>` inside the stateful CLI to see the flags supported by your installed version.

3.2 Process specification grammar

Process specifications accept `to` or `>` between space-separated initial and final states. Braces express alternatives in a cross-section final state, while `{}` denotes an empty state:

```
e+ e- > mu+ mu-
e+ e- > { Z Z, a a, H H }
{} > {}
```

A slash vetoes particles and a vertical bar selects an allow-list. The words `veto` and `only` are equivalent forms. Amplitude coupling constraints use the coupling name directly; powered constraints apply to the complete cross section:

```
e+ e- > d d~ g / u c QED==2 QCD>=2 QCD<=4
e+ e- > mu+ mu- | g ghG QED^2==2 QCD^2>=2
```

The bracketed perturbative block is intentionally distinct from ordinary flags. `{n}` fixes the amplitude loop count (or the sum across the two cut sides), `{{n}}` fixes the forward-graph loop count, and `QCD=n` or the shorthand `QCD` selects a relative perturbative order:

```
e+ e- > Z [ {1} {{2}} QCD=2 QED=1 ]
e+ e- > Z [ QCD ]
```

Resolve names through the active model

Particle names, antiparticles, coupling-order names, and allowed interactions come from the model loaded in the current state. A syntactically valid specification can still be rejected when that model does not define a requested name.

3.3 Filters, grouping, and multiplicity

Generation flags control topology filters, loop ranges, cut blobs and spectators, symmetrization, and numerator-aware grouping. These choices are physics-significant. In particular, grouping can combine graph multiplicities and numerator ratios; external-fermion symmetrization requires the caller to retain the associated ordering signs. Run `generate help <mode>` before choosing filters to see the available aliases, defaults, values, and argument counts.

The generated graph's overall factor keeps separate contributions for automorphisms, internal fermion loops, external-fermion ordering, numerator-independent symmetry grouping, and numerator-dependent grouping. Persist the state and inspect that factor when auditing diagram normalization instead of replacing it with an assumed factorial.

3.4 A maintainable generation card

The following reduced card uses the repository's scalar model and the same one-loop bubble command exercised by the maintained scalar-topology example. Save it as `manual-bubble.toml` and run `./gamma-loop --clean-state manual-bubble.toml run generate -c "quit -o"` from the repository root.

```
commands = []

[cli_settings.state]
folder = "./gamma-loop-state/manual-bubble"
name = "manual_bubble"

[[command_blocks]]
name = "generate"
commands = [
  "import model scalars-default.json",
  "generate amp scalar_1 > scalar_1 [{1}] --allowed-vertex-interactions V_3_SCALAR_122 -p bubble -i scalar_bubble",
  "generate",
  "save state -o",
]
```

The acceptance invariant is a persisted state in which `display` processes identifies process bubble and integrand scalar_bubble; replaying the card with a clean destination must produce the same named objects. The exact command and setting semantics are linked from the amplitude-generation reference, state-persistence reference, and state-folder setting. The full maintained source is the scalar bubble run card.

Interpret the first failing stage

An unknown particle or `V_3_SCALAR_122` error means the scalar model was not imported or no longer exposes that interaction. A process with no retained diagram points to the process specification or generation filters. A generated process that disappears after restart means `save state -o` did not complete or the resumed `-s` path differs from `cli_settings.state.folder`.

3.5 From generation to evaluation

A reproducible run keeps its stages explicit:

- import or select the model before resolving a process specification;
- generate the process and inspect the retained diagrams;
- generate the requested integrand representation;
- set kinematics, sampling, stability, and subtraction settings;
- evaluate samples or run an integration command block;
- exit with persistence enabled when the state should be resumed.

3.6 Select integration slots and targets

An integration slot is one (process, integrand) pair. Repeat `-p/--process` and `-i/--integrand-name` in matching order to integrate several slots together. By default those slots train and sample a shared grid, so their sample shapes must be compatible; add `--uncorrelated` when each slot needs an independent grid.

A single `--target re im` (or `--target re,im`) applies the same known comparison value to every selected slot. Multi-slot runs can instead use one keyed value per slot:

```
integrate \  
-p bubble_no_integrated_UV -i scalar_bubble_below_thres \  
-p bubble -i scalar_bubble_below_thres \  
--target bubble_no_integrated_UV@scalar_bubble_below_thres=1.471664021721597e-2,0.0 \  
      bubble@scalar_bubble_below_thres=2.7029875684552542e-3,0.0 \  
--workspace-path ./integration/bubble-below-threshold \  
--n-cores 1
```

Targets annotate result deltas; they do not replace `integrator.target_relative_accuracy`, `integrator.target_absolute_accuracy`, or the sample-budget settings that control convergence. The maintained `scalar bubble` run card contains the complete settings and both below- and above-threshold target sets. The generated integration reference is authoritative for selectors, renderers, batching controls, and allowed values.

3.7 Resume or replace an integration workspace

An integration workspace is a completed-iteration checkpoint, not a second `GammaLoop` state. Without `--workspace-path`, writable sessions use `STATE_FOLDER/integration_workspace`; read-only sessions use `./integration_workspace_STATE_NAME` (or `./integration_workspace`) so they do not write into the active state.

- Without `--restart`, `GammaLoop` resumes when the workspace already contains a compatible checkpoint. Slots, shared-versus-uncorrelated sampling, effective model parameters, and generated-integrand fingerprints must match. Saved targets and non-model runtime settings win over changed values and produce a warning.
- With `--restart`, `GammaLoop` removes that specific workspace and starts the integration from scratch. Use it when changing slots, sampling correlation, generated integrands, or settings that should take effect. It does not mean “resume”.
- `--show-summary-only` reads the last completed result without evaluating more samples and cannot be combined with `--restart`.

The workspace root contains `manifest.json`, the latest user-facing `integration_result.json`, per-slot settings and results under `integrands/`, and the internal resume checkpoint under `state/integration_state.bin`. Observable resume snapshots live at `state/observables/<process@integrand>/latest.json`. Configured user-facing output is written per slot as `integrands/<process@integrand>/observables_final.json` or `integrands/<process@integrand>/observables_final.hwu`. `--write-results-for-each-iteration` additionally retains numbered result and observable history at the corresponding root or slot boundary.

Archive the right boundary

`integration_result.json` is a presentation snapshot, not the authoritative resume state. Keep the complete

integration workspace when a run must be resumed, and keep the GammaLoop state/run card that produced the referenced integrands. A copied result JSON alone is suitable for analysis, not continuation.

Keep the complete state together

`run.toml` records boot settings, reusable command blocks, and top-level commands. Generated process and integrand artifacts live beside it. Resume with `-s STATE_FOLDER`; replay the run card to reconstruct the state from scratch. Copy or archive the entire state directory rather than moving a generated subdirectory on its own, because settings and fingerprints are checked together.

Contributor-facing ownership and data flow are documented in the [GammaLoop architecture](#) and the [UV renormalization architecture](#).

4 Inspect events and observables

GammaLoop can evaluate selected points without launching a full adaptive integration. This is the shortest path for inspecting the generated cut events, testing selectors, and checking that histograms receive the expected weights. It is an investigative workflow: a few chosen points cannot establish convergence or replace the persisted result of an integration.

4.1 Start from the maintained differential example

The repository contains matching Python and Rust journeys for the differential Local Unitarity process $e^+ e^- \rightarrow d \bar{d} g$. Their run cards generate the real two-graph process and configure:

- event generation and additional event weights;
- jet transverse-momentum, jet-count, and down-quark-energy quantities;
- a leading-jet selector; and
- continuous and discrete histogram observables.

The run card's `display_named_settings_examples` block lets you inspect each named object before evaluation:

```
display quantities -p #0
display quantities -p #0 leading_jet_pt
display selectors -p #0 leading_jet_pt_cut
display observables -p #0 leading_jet_pt_hist
```

Use these displays as a preflight check. A missing quantity name belongs to configuration; an empty event list belongs to event generation or selection; an unexpected histogram belongs to the quantity, phase, binning, or weights. Keeping those stages separate is more informative than debugging the final number alone.

4.2 Run the Python journey

Build the native package, then run the maintained script from the repository root. It creates a fresh state beside the script, generates the configured process, evaluates one integration-space point, evaluates a two-point batch, and evaluates one momentum-space point:

```
nix develop
uv venv .venv
source .venv/bin/activate
just build-api
python examples/api/python/epem_a_ddxg_xs_L0/inspect_events.py
```

The exact workflow is kept in the Python event-inspection script and its [differential run card](#). The core API sequence is:


```

from pathlib import Path

import numpy as np
from gammaloop import GammaLoopAPI

example = Path("examples/api/python/epem_a_ddxg_xs_L0")
api = GammaLoopAPI(
    state_folder=example / "state",
    boot_commands_path=example / "run.toml",
    clean_state=True,
)
api.run("run display_named_settings_examples")

point = [0.17, 0.31, 0.53, 0.23, 0.41, 0.67]
single = api.evaluate_sample(point, return_events=True)

points = np.array([
    point,
    [0.11, 0.29, 0.47, 0.19, 0.37, 0.59],
], dtype=float)
batch = api.evaluate_samples(points, return_events=True)

print(single.integrand_result, single.generated_event_count)
for group in single.event_groups:
    for event in group.events:
        print(event.cut_info.graph_id, event.weight, event.outgoing_momenta)
print(batch.observables["leading_jet_pt_hist"].sample_count)

```

`return_events=True` temporarily requests returned events for that call; it does not permanently rewrite the integrand setting. Every `EventGroup` contains correlated accepted events produced by one graph group. Each event carries incoming/outgoing momenta, its primary complex weight, cut and graph identifiers, and any configured additional weights.

Single and batch results have different aggregation boundaries

`evaluate_sample` returns one `EvaluationResult`: its event groups and observable snapshot belong to that sample. `evaluate_samples` returns per-row `SampleEvaluationResult` objects and one `BatchEvaluationResult.observables` dictionary merged across the whole batch. Do not read a batch histogram as if it belonged to the last row, and do not add the same batch snapshot once per sample.

The histogram values are immutable snapshots. Inspect bins, underflow/overflow, raw sums of weights and squared weights, entry counts, and the aggregate statistics before deriving a plot. Keep raw statistics when combining results; rounded rendered values are not a merge format.

4.3 Run the Rust journey

The Rust example loads the same kind of run card through `StateLoadOption`, executes the display block through a CLI session, then calls the typed evaluation functions. It is a `rust-script` program so its dependency revision is recorded with the example:

```

NO_SYMBOLICA_OEM_LICENSE=1 \
EXTRA_MACOS_LIBS_FOR_GNU_GCC=T \
SYMBOLICA_LICENSE="$SYMBOLICA_LICENSE" \
rust-script --debug examples/api/rust/epem_a_ddxg_xs_L0/inspect_events.rs

```

The documentation gate compiles the request boundary below against the workspace API. It keeps the event-return override and every required field synchronized even though the licensed process evaluation remains outside the pull-request tier:


```

use gammaloop_api::commands::evaluate_samples::EvaluateSamples;
use ndarray::arr2;

let points = arr2(&[[0.17, 0.31, 0.53, 0.23, 0.41, 0.67]]);
let request = EvaluateSamples {
    process_id: None,
    integrand_name: None,
    use_arb_prec: false,
    minimal_output: false,
    return_generated_events: Some(true),
    momentum_space: false,
    points: points.view(),
    integrator_weights: None,
    discrete_dims: None,
    graph_names: None,
    orientations: None,
};
assert_eq!(request.return_generated_events, Some(true));

```

Read the [Rust event-inspection script](#) with its [run card](#). The ordinary `EvaluateSamples` path returns the cross-language f64 boundary. Rust callers that must retain the active numeric precision use `EvaluateSamplesPrecise`; arbitrary-precision internal evaluation does not make the ordinary result fields arbitrary-precision.

Verification and resource cost

The pull-request gate parses the commands, compiles the Python source, and type-checks the Rust request literal, but does not run process generation or native evaluation. The maintained scripts are the real acceptance path and require a built API, the repository's scientific dependencies, and an appropriate Symbolica license. They create or replace only their example-local state directory because they opt into `clean_state=True`; remove that option before pointing at work you need to keep.

4.4 From point inspection to integration output

Selected points answer structural questions: whether events are produced, how graph/cut metadata is grouped, which selector accepts them, and which histogram bins are filled. A production result still needs an integration workspace, target-accuracy settings, iteration diagnostics, and the final observable output. Continue with the [state and integration-workspace guide](#) for resume semantics and the distinction between `integration_result.json`, `observables_final.*`, and the authoritative checkpoint.

Contributors changing progress reporting, cancellation, or terminal ownership should also read the [integration dashboard architecture](#).

The exact Python return types and fields are linked from the [interface guide](#) and generated Python API.

5 Logging and diagnostics

This guide explains the default tracing configuration used by the CLI and API at startup and gives copyable filters for narrowing a noisy calculation to the subsystem and work unit that matter.

Before you start

Prerequisites: a built GammaLoop CLI and a state or run card that reaches the behavior you want to inspect. Filter changes take effect immediately and add negligible setup time, but verbose `inspect` or `dump` events can produce large files and slow a run. The expected invariant is that a narrow directive changes diagnostic visibility, not numerical results. If a filter produces nothing, first use a broader target at `debug`, confirm that release compilation has not removed the callsite, and then add tag constraints. After isolating the event,

record the exact directive with the run card and continue to the CLI/settings reference for persistence and state controls.

5.1 Default directives

Normal session defaults come from GlobalSettings:

- `display_directive = "info"`
- `logfile_directive = "off"`

These are the application defaults written into `global_settings.toml` when defaults are shown or persisted.

In practice, `info` is the current convention for normal screen-facing output. Selective internal diagnostics should generally stay on debug and be turned on via narrower target/tag filters rather than by raising the whole display logger.

5.2 Startup precedence

Startup applies logging in this order:

1. Base settings come from GlobalSettings.
2. Environment overrides replace the base spec:
 - `GL_ALL_LOG_FILTER` overrides both stderr and logfile filters.
 - `GL_DISPLAY_FILTER` overrides only the stderr/display filter.
 - `GL_LOGFILE_FILTER` overrides only the logfile filter.
3. CLI overrides are then applied:
 - `-l/--level` overrides the stderr/display filter for the session.
 - `-L/--logfile-level` overrides the logfile filter for the session.
4. Some CLI modes hard-disable logfile logging for the session:
 - `--read-only-state`
 - `--logfile-level off`

When logfile logging is hard-disabled at boot, later settings changes cannot re-enable it for that session.

5.3 CLI level mapping

`-l/--level` and `-L/--logfile-level` expand to explicit crate directives:

- `off` -> `gammaloop_api=off,gammalooprs=off`
- `error` -> `gammaloop_api=error,gammalooprs=error`
- `warn` -> `gammaloop_api=warn,gammalooprs=warn`
- `info` -> `gammaloop_api=info,gammalooprs=info`
- `debug` -> `gammaloop_api=debug,gammalooprs=debug`
- `trace` -> `gammaloop_api=trace,gammalooprs=trace`

5.4 Empty-spec fallback floors

The filter builders also have hardcoded fallback floors used when a spec is empty:

- display/stderr builder fallback: `off`
- logfile builder fallback: `warn`

This is separate from the normal application defaults above. In other words:

- if startup uses the normal settings path, the defaults are `info` for display and `off` for logfile
- if an empty filter string is explicitly parsed, the display builder falls back to `off` and the logfile builder falls back to `warn`

5.5 Runtime updates

Changing `global.display_directive` or `global.logfile_directive` after startup reloads the active tracing filters. The CLI `stderr` override from `-l/--level` is treated as a session override and is separate from the persisted global setting.

5.6 Tag-based debug filtering

For selective debug logging, GammaLoop uses its own logging DSL rather than raw `EnvFilter`.

This follows the general idea in Tom Mrazik's [tag-based logging note](#).

The main reason for owning the DSL is that GammaLoop needs composable tag groups and stable semantics around presence, absence, and explicit boolean values.

- Use stable targets for broad subsystems such as `gammaloops::uv::forest` or `gammaloops::integrands::process::cross_section`.
- Add boolean event fields as composable tags.
- Filter by field presence with `target[{tag_a,tag_b,!tag_c}]=debug`, or by displayed boolean values with `target[{#tag_a,#!tag_c}]=debug`.

This is preferable to span-name filtering because span-based filters can admit child-library events emitted while the span is active. GammaLoop tag filters only look at the event target and the boolean tag fields on the event itself.

5.6.1 Supported directive syntax

The supported directive forms are:

- `level`
- `target=level`
- `target[{tag_a,tag_b,!tag_c}]=level`
- `target[{#tag_a,#!tag_c}]=level`
- `target`

Notes:

- Directives are comma-separated.
- `target` is matched as a module-style prefix, so `gammaloops::uv=debug` matches `gammaloops::uv` and `gammaloops::uv::forest`.
- A directive without an explicit `=level` is treated as `=trace`.
- Tag groups are conjunctions: `[{generation,uv}]` means both tags must match.
- `!tag` means the field is absent.
- `#tag` means the field is present with boolean value `true`.
- `#!tag` means the field is present with boolean value `false`.
- `field=value` matches an explicit field value.

5.6.2 Tag matching model

Tags are represented by event fields.

- `#generation` at the callsite emits `generation = true`.
- In a directive, `generation` means the event must carry a field named `generation`.
- In a directive, `#generation` means the event must carry `generation = true`.
- In a directive, `!inspect` means the event must not carry a field named `inspect`.
- In a directive, `#!inspect` means the event must carry `inspect = false`.
- In a directive, `inspect=true`, `mode=summary`, or `label="soft region"` means the event must carry that value specifically.

This means the callsite controls both:

- whether a log is selectable by a presence/absence query
- whether a log can be selected by an explicit value query such as `inspect=false` or `mode=summary`

5.6.3 Static vs dynamic filtering

Presence/absence-only directives stay on the lazy metadata path.

- `generation`
- `!inspect`
- `gammaloops::uv::forest[{generation,uv,!dump}]=debug`

These can be decided from the callsite target and declared field names alone, so the filter can answer always or never at callsite registration time.

Value-matching directives are dynamic.

- `inspect=false`
- `#!inspect`
- `#generation`
- `mode=summary`
- `label="soft region"`

Those depend on the event instance, so they are evaluated in the per-event pass.

5.6.4 Precedence

When multiple directives match the same event, GammaLoop prefers the most specific one:

- longer target prefix wins
- then the directive with more tag requirements wins
- then later directives win if the earlier specificity is identical

This lets broad directives such as `gammaloops=info` coexist with narrow tag-specific directives such as `gammaloops::uv::forest[{generation,uv,dump}]=debug`.

5.6.5 Copy-pasteable display tags

Display logs render boolean fields next to the source as a directive tag group:

```
@gammaloops::uv::forest[{#generation, #uv, #!inspect}] DEBUG: message
```

Every boolean field is rendered this way, not only the standard tag vocabulary. `true` becomes `#field_name`; `false` becomes `#!field_name`. These boolean fields are removed from the normal display field table.

Use the `full` logging prefix when you want the rendered source to be a valid directive head:

```
gammaloop --logging-prefix full -l debug ...
```

Then the text after `@` and before the log level can be copied into a directive by adding `=debug`, for example:

```
display_directive = "gammaloops::uv::forest[{#generation, #uv, #!inspect}]=debug"
```

This copy-paste form assumes the display source is the module target. Enabling `full_line_source` adds file and line information to the source display, which is useful for locating code but is not a valid directive target.

5.6.6 Sink-only field prefixes

This repo also has sink-routing prefixes for fields:

- `file.<name>`: only rendered into the logfile/json sink

- `display.<name>`: only rendered into the `stderr/display` sink

Examples:

- `file.integrands = %...` will appear only in the file logger output
- `display.progress = %...` will appear only in the display logger output

These prefixes are a formatting/routing feature, not a separate event kind. They decide where a field is rendered after the event has been emitted. This lets a callsite attach large payloads such as `file.integrands` without dumping them to `stderr`.

5.6.7 Pipeline-wide tag set

Prefer a small stable vocabulary that cuts across the whole pipeline.

Primary phase tags:

- `generation`: work that builds or prepares runtime objects before Monte Carlo integration starts
- `integration`: work performed while evaluating samples or managing the adaptive integration loop
- `profile`: UV/IR/profile-style diagnostic scans and their analysis
- `persistence`: reading or writing saved state, manifests, checkpoints, and exported results

Core domain tags:

- `uv`: ultraviolet counterterms, UV forests, UV profiles, or UV-specific generation/evaluation logic
- `ir`: infrared subtraction, IR profiles, or IR-specific generation/evaluation logic
- `subtraction`: threshold, UV, or IR subtraction logic when the main concern is subtraction rather than the specific regime
- `sampling`: channel choice, discrete axes, parameterizations, or sample-generation choices
- `stability`: precision escalation, retries, instability diagnosis, and numerical safety checks
- `observables`: histogramming, event-to-observable projection, and observable snapshot production
- `selectors`: event-selection logic and selector decisions
- `cache`: cache lookup, reuse, invalidation, or cache-debug instrumentation

Common work-unit tags:

- `graph`: the log is about one graph or graph-local data
- `group`: the log is about a graph group or another explicitly grouped aggregate
- `orientation`: the log is about orientation-dependent data or choosing/summing orientations
- `channel`: the log is about multi-channeling or a particular channel
- `cut`: the log is about a cut, raised cut, or cut-local computation
- `event`: the log is about generated/retained event objects or event-group processing
- `sample`: the log is about one evaluation sample, its coordinates, or per-sample intermediate values
- `iteration`: the log is about one adaptive integration iteration or iteration-level summaries
- `term`: the log is about one algebraic term, summand, or term-local contribution

Common purpose tags:

- `solver`: the log is about root-finding, linear solves, fitting, or similar numerical solver state
- `compile`: the log is about evaluator/code generation or compilation-oriented preparation
- `inspect`: the log exposes detailed intermediate state for debugging, rather than a high-level milestone
- `summary`: the log is a compact roll-up rather than a step-by-step trace
- `dump`: the log emits large or structured payloads such as expressions, tables, or serialized views

5.6.8 Tag boundaries

Use the most general tag that accurately captures the reason you want to turn the log on or off.

- Prefer `generation` over `compile` when the message is a broad generation milestone.
- Add `compile` only when the message is specifically about evaluator construction or compilation-like work.
- Prefer `uv` or `ir` when the regime matters to the user; use `subtraction` when the subtraction mechanism is the real concern.

- Prefer `inspect` for verbose intermediate values that are mainly useful while debugging internals.
- Add `dump` when the payload is large enough that users may want to suppress it separately from lighter debug logs.
- Do not use `graph`, `cut`, or `sample` as substitutes for generation or integration; they refine phase tags rather than replace them.

5.6.9 Naming guidance

- Use phase tags first. They answer “when in the pipeline did this happen?”
- Use domain tags second. They answer “what subsystem is this about?”
- Use work-unit tags third. They answer “what object is being worked on?”
- Use purpose tags last for optional refinement.
- Prefer tags that are meaningful across multiple modules.
- Avoid tags that merely restate a function name.
- Avoid tags tied to one internal representation unless that representation is a stable concept in user-facing debugging.

`parametric` is usually not a core pipeline tag. It is acceptable as a local refinement when needed, but should not be treated as part of the primary vocabulary unless it becomes a consistently useful cross-cutting concept.

5.6.10 Example queries

- all generation-time UV forest logs:
 - `gammaloops::uv::forest[{generation,uv}]=debug`
- only the large UV forest dumps for orientation-dependent generation logs:
 - `gammaloops::uv::forest[{generation,uv,orientation,dump}]=debug`
- UV forest term-by-term generation logs:
 - `gammaloops::uv::forest[{generation,uv,term}]=debug`
- all integration-time subtraction debug in amplitude evaluation:
 - `gammaloops::integrands::process::amplitude[{integration,subtraction}]=debug`
- per-sample integration inspection in cross-section evaluation:
 - `gammaloops::integrands::process::cross_section[{integration,sample,inspect}]=debug`
- all integration-time cut solver debug in cross-section evaluation:
 - `gammaloops::integrands::process::cross_section[{integration,cut,solver}]=debug`
- integration-time debug excluding inspect-tagged logs:
 - `gammaloops::integrands::process::cross_section[{integration,!inspect}]=debug`
- only events that explicitly set `inspect=true`:
 - `gammaloops::integrands::process::cross_section[{integration,inspect=true}]=debug`
- only events with a specific textual mode field:
 - `gammaloops::integrands::process::cross_section[{integration,mode=summary}]=debug`

Example display directive:

```
[cli_settings.global]
display_directive =
"gamma_loop_api=info,gammaloops=info,symbolica=off,poly::gcd=off,gammaloops::uv::forest[{generation,uv,orientation,dump}]=debug"
```

5.7 Unsupported EnvFilter syntax

GammaLoop no longer treats the directive string as generic `EnvFilter` syntax.

In particular, do not rely on:

- span-name filters such as `target[span_name]=debug`
- generic field filters such as `[{field=value}]`
- `EnvFilter` regex semantics

If those are needed later, they must be added explicitly to the GammaLoop DSL.

5.8 Release-build note

gammaloops is compiled with tracing feature `release_max_level_info`, so `debug!` and `trace!` callsites in that crate are compiled out in release builds.

5.9 Authoritative implementation

The startup precedence and sink wiring live in the [API tracing setup](#). The shared directive parser, CLI level mapping, and filter behavior live in the [runtime tracing utilities](#), while persisted defaults are owned by `GlobalSettings`.

6 Kurvst curve and path API

Kurvst is GammaLoop's Typst-facing WebAssembly layer for Bezier and Hobby paths, arc-length trimming, sampled path patterns, parallel curves, and CeTZ conversion. This page embeds the complete maintained Kurvst manual used by the drawing pipeline.

Geometry rather than topology

Kurvst transforms curve geometry. Linnest and Linnest own graph topology and layout; GammaLoop's drawing templates combine those results with Kurvst paths and styles.

kurvst is a Typst package backed by `kurvst.wasm`, a small Kurbo-based plugin. Its public API is path-based: construct a path dictionary, pass that path into transforms, and draw the returned path.

It exposes Bezier utilities that are awkward or unavailable in native Typst:

- constructing cubic and Hobby paths,
- trimming paths by arc length,
- generating sampled path patterns,
- fitting Kurbo offset/parallel paths,
- converting returned path geometry to CeTZ drawing commands.

6.1 Choose an import path

Kurvst is currently a bundled source package, not a Typst Universe package. A Clinnet run writes it below `build/templates/crates/kurvst/typst/`; a custom template in `build/templates/` can therefore import it with:

```
#import "crates/kurvst/typst/src/lib.typ" as kurvst
```

Repository examples live one directory beside `src/` and instead use `#import "../src/lib.typ" as kurvst`. Keep `kurvst.wasm`, `typst.toml`, and the `src/` tree together when copying the package to another Typst project.

Path geometry points are two-item tuples:

```
#let p = (1.2, -0.4)
```

Cubic segments use `start`, `control-start`, `control-end`, and `end`:

```
#let segment = (  
  start: (0, 0),  
  control-start: (1, 0.5),  
  control-end: (2, -0.5),  
  end: (3, 0),  
)
```


Use `from-cubic(segment)` once to turn that cubic dictionary into a path dictionary. Returned paths contain a curve-command wire path in `path`. Rust converts that wire path to Kurbo's `BezPath` internally, and every path-level Kurvst function accepts either a returned dictionary or the raw wire path.

```
#let path = kurvst.from-cubic(segment)
#let trimmed = kurvst.trim(path, start-outset: 0.2, end-outset: 0.1)
#let shifted = kurvst.parallel(trimmed, distance: 0.15)
```

6.2 Path Wire Format

A Kurvst wire path is a dictionary with one `elements` array. The elements mirror Typst's native curve commands, but points are plain numeric tuples:

```
#let path = (
  elements: (
    (kind: "move", start: (0, 0)),
    (kind: "line", end: (0.6, 0.3)),
    (kind: "quad", control: (1.0, 1.0), end: (1.4, 0.3)),
    (
      kind: "cubic",
      control-start: (1.8, -0.4),
      control-end: (2.4, 1.0),
      end: (3.0, 0),
    ),
    (kind: "close", mode: "straight"),
  ),
)
```

The supported element kinds are:

- `move`: starts a subpath at `start`.
- `line`: adds a straight segment ending at `end`.
- `quad`: adds a quadratic segment using `control` and `end`.
- `cubic`: adds a cubic segment using `control-start`, `control-end`, and `end`.
- `close`: closes the current subpath. `mode` is optional and defaults to `"straight"` when emitted by Kurvst.

Prefer the path fragment helpers for new code. `line`, `quad`, and `cubic` include their start points, so fragments can be built independently:

```
#let custom = kurvst.path(
  kurvst.line((0, 0), (0.6, 0.3)),
  kurvst.quad((0.6, 0.3), (1.0, 1.0), (1.4, 0.3)),
  kurvst.cubic((1.4, 0.3), (1.8, -0.4), (2.4, 1.0), (3.0, 0)),
)
```

`path` and `append` flatten fragments. If an appended fragment starts where the current path ends, its leading move is skipped; if it starts elsewhere, the move begins a new subpath. The edited path can go straight back into Kurvst transforms:

```
#let base = kurvst.from-cubic(segment)
#let extended = kurvst.append(base, kurvst.line(segment.end, (3.4, 0.2)))

#let trimmed = kurvst.trim(extended, start-outset: 0.15)
#let shifted = kurvst.parallel(trimmed, distance: 0.1)

#kurvst.to-native(shifted, unit: 36pt, stroke: rgb("#1b7f4c") + 0.7pt)
```

Use `elements` when you need the raw command stream, `points` for the visited endpoints, `segments` for cubic segment dictionaries, `to-native` for native Typst drawing, and `to-cetz-data` or `to-cetz` for CeTZ.

A typical chain keeps the returned dictionaries intact until the final drawing step:

```
#let base = kurvst.hobby-through(demo-start, demo-through, demo-end)
#let trimmed = kurvst.trim(base, start-outset: 0.2, end-outset: 0.1)
#let shifted = kurvst.parallel(trimmed, distance: 0.15)

#cetx.canvas({
  kurvst.to-cetz(base, stroke: rgb("#bbbbbb") + 0.4pt)
  kurvst.to-cetz(trimmed, stroke: rgb("#d72638") + 0.6pt)
  kurvst.to-cetz(shifted, stroke: rgb("#1b7f4c") + 0.6pt)
})
```

6.3 Paths And Trimming

`cubic` constructs a path from one cubic Bezier. `trim` trims by curve distance from the start and/or end and returns another path dictionary.

```
#let path = kurvst.from-cubic(segment)
#let trimmed = kurvst.trim(path, start-outset: 0.15, end-outset: 0.1)
```

`hobby-through(start, through, end)` builds a smooth open curve through three points; `segments` returns the two cubic halves split at `through`. `hobby-spline(points)` does the same for any open point sequence with at least two points. Both return path dictionaries and can be passed directly to path-level helpers.

```
#let spline = kurvst.hobby-spline((
  (0.0, 0.0),
  (0.9, 0.8),
  (1.8, -0.3),
  (2.8, 0.4),
))
#let shifted = kurvst.parallel(spline, distance: 0.14)
#let wiggle = kurvst.pattern(
  spline,
  pattern: kurvst.wave(samples-per-period: 24),
  amplitude: 0.1,
  wavelength: 0.55,
)
#cetx.canvas({
  kurvst.to-cetz(spline, stroke: rgb("#bbbbbb") + 0.45pt)
  kurvst.to-cetz(shifted, stroke: rgb("#1b7f4c") + 0.6pt)
  kurvst.to-cetz(wiggle, stroke: rgb("#355c9a") + 0.55pt)
})
```

`split-through` combines Hobby spline construction with endpoint trimming and returns one path part between each consecutive point. Consumers such as Linnest can wrap it for graph-edge geometry.

6.4 Path Patterns

`pattern` generates a one-dimensional pattern along any path dictionary. Built-ins are regular Typst dictionaries:

```
#let wave = kurvst.wave()
#let zigzag = kurvst.zigzag()
#let coil = kurvst.coil(longitudinal-scale: 1.6)
```

String names "wave", "zigzag", and "coil" are accepted for convenience, but they are resolved in Typst before calling wasm.

Custom patterns use the same object shape:

```
#let hook = (  
  kind: "points",  
  name: "hook",  
  interpolation: "linear",  
  points: (  
    (at: 0, x: 0, y: 0),  
    (at: 0.25, x: 0.15, y: 1),  
    (at: 0.5, x: 0, y: 0),  
    (at: 0.75, x: -0.15, y: -1),  
    (at: 1, x: 0, y: 0),  
  ),  
)
```

at is the phase position inside one wavelength, from 0 to 1. The x coordinate offsets along the path tangent and y offsets along the path normal; both are scaled by amplitude. If at is omitted, points are spaced evenly. Use interpolation: "linear" for corners and "smooth" for a spline through sampled points.

endpoint-ramp: true tapers amplitude at anchored endpoints. This is useful for coils, whose longitudinal offset would otherwise put the first and last visible points inside the turn near nodes.

```
#let base = kurvst.from-cubic(segment)  
#let path = kurvst.pattern(  
  base,  
  pattern: hook,  
  amplitude: 0.15,  
  wavelength: 0.7,  
)  
#native-scene({  
  native-cubics(kurvst.segments(base), stroke: rgb("#c8c8c8") + 0.45pt)  
  native-polyline(kurvst.points(path), stroke: rgb("#1b7f4c") + 0.8pt)  
})
```

6.5 Parallel Paths

parallel uses Kurbo's offset curve fitter to produce a path at a fixed normal distance from the source path. Positive distances follow the left normal of the path direction; negative distances follow the right normal. start-outset and end-outset trim the fitted path by arc length after the offset is computed.

```
#let base = kurvst.from-cubic(segment)  
#let left = kurvst.parallel(base, distance: 0.18, start-outset: 0.35, end-outset: 0.35)  
#let right = kurvst.parallel(base, distance: -0.18)  
#native-scene({  
  native-cubics(kurvst.segments(base), stroke: rgb("#c8c8c8") + 0.45pt)  
  native-cubics(kurvst.segments(left), stroke: rgb("#1b7f4c") + 0.8pt)  
  native-cubics(kurvst.segments(right), stroke: rgb("#355c9a") + 0.8pt)  
})
```

6.6 Path Layers

layer is a convenience wrapper for drawing derived visible paths. It combines side-aware offsetting, endpoint outsets, and centered shortening into one path-in/path-out operation. length is a fixed visible arc length, ratio is a fraction of the input path length, and resolve-length decides how to combine them. The default "min" keeps whichever limit is shorter.

```
#let base = kurvst.hobby-spline((
  (0.0, 0.0),
  (0.9, 0.8),
  (1.8, -0.3),
  (2.8, 0.4),
))
#let layer = kurvst.layer(
  base,
  offset: 0.16,
  length: 1.6,
  ratio: 0.5,
  resolve-length: "min",
)
#cetz.canvas({
  kurvst.to-cetz(base, stroke: rgb("#bbbbbb") + 0.45pt)
  kurvst.to-cetz(layer, stroke: rgb("#1b7f4c") + 0.8pt)
})
```

Use side-point to choose the sign of the offset from a point on the desired side of the path. Drawing packages can use this to place derived layers on the same side as an edge label without knowing anything about the physics or graph style that requested the layer.

6.7 Native Drawing Primitives

Kurvst returns dictionaries and arrays. The core geometry can be drawn without CeTZ by emitting native curve content:

```
#kurvst.to-native(kurvst.from-cubic(segment),
unit: 36pt, stroke: black + 0.7pt)
```



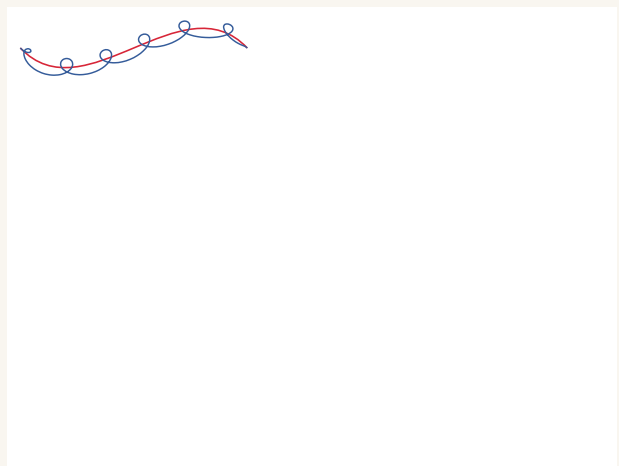
The generated reference below keeps drawing code inline. It only uses the small common helpers above for coordinate scaling, native curve construction, point markers, and fixed-size preview blocks.

6.8 CeTZ Interoperability

The CeTZ helpers are thin adapters over the same returned geometry. Use them when the surrounding document already lives in a CeTZ canvas, or when you want CeTZ path merging and styling:

```
#cetx.canvas({
  let base = kurvst.from-cubic(segment)
  kurvst.to-cetz(base, stroke: rgb("#d72638")
+ 0.6pt)

  let path = kurvst.pattern(
    base,
    pattern: kurvst.coil(longitudinal-scale:
1.6),
    amplitude: 0.12,
    wavelength: 0.55,
  )
  kurvst.to-cetz(path, stroke: rgb("#355c9a")
+ 0.55pt)
})
```



6.9 Generated Reference

6.10 kurvst

- `append()`
- `center-outset()`
- `close()`
- `coil()`
- `cubic()`
- `cubic-point()`
- `cubic-tangent()`
- `cubic-to()`
- `elements()`
- `from-cubic()`
- `from-elements()`
- `hobby-spline()`
- `hobby-through()`
- `layer()`
- `length()`
- `line()`
- `line-segment()`
- `line-to()`
- `move-to()`
- `outset-point()`
- `parallel()`
- `path()`
- `pattern()`
- `point()`
- `points()`
- `quad()`
- `quad-to()`
- `resolve-length()`
- `segments()`
- `split-through()`
- `to-cetz()`
- `to-cetz-data()`
- `to-native()`
- `trim()`
- `wave()`
- `zigzag()`

Variables

- `layer-defaults`

6.10.1 append

Return a path with additional fragments or elements appended.

6.10.1.1 Parameters

```
append(  
  path: dictionary ,  
  ..parts: any  
) -> dictionary
```

6.10.1.1.1 path dictionary

Base Kurvst path dictionary.

6.10.1.1.2 ..parts any

Path fragments, path elements, or element arrays to append.

6.10.2 center-outset

Compute the symmetric trim needed to center a shorter path layer.

6.10.2.1 Parameters

```
center-outset(  
  base-length: int float ,  
  length: none int float ,  
  ratio: none int float ,  
  resolve-length: string function ,  
  start-outset: int float ,  
  end-outset: int float  
) -> int float
```

6.10.2.1.1 base-length int or float

Full base path arc length.

6.10.2.1.2 length none or int or float

Fixed target visible length.

Default: none

6.10.2.1.3 ratio none or int or float

Relative target visible length as a fraction of base-length.

Default: none

6.10.2.1.4 resolve-length string or function

Resolution strategy for fixed and relative targets.

Default: "min"

6.10.2.1.5 start-outset `int` or `float`

Already-applied trim at the start of the path.

Default: `0`

6.10.2.1.6 end-outset `int` or `float`

Already-applied trim at the end of the path.

Default: `0`

6.10.3 close

Build a close path element.

6.10.3.1 Parameters

```
close(mode: string) -> dictionary
```

6.10.3.1.1 mode `string`

Native Typst curve close mode.

Default: `"straight"`

6.10.4 coil

A smooth coil path pattern.

6.10.4.1 Parameters

```
coil(  
  samples-per-period: int,  
  longitudinal-scale: int float  
) -> dictionary
```

6.10.4.1.1 samples-per-period `int`

Number of samples used to approximate one coil period.

Default: `16`

6.10.4.1.2 longitudinal-scale `int` or `float`

Horizontal scale of the coil before it is mapped onto a path.

Default: `1.25`

6.10.5 cubic

Build a cubic path fragment.

6.10.5.1 Parameters

```
cubic(  
    start: array ,  
    control-start: array ,  
    control-end: array ,  
    end: array  
) -> dictionary
```

6.10.5.1.1 start array

Start point.

6.10.5.1.2 control-start array

Cubic control point near start.

6.10.5.1.3 control-end array

Cubic control point near end.

6.10.5.1.4 end array

Endpoint.

6.10.6 cubic-point

Evaluate a cubic segment at parameter t.

6.10.6.1 Parameters

```
cubic-point(  
    segment: dictionary ,  
    t: int float  
) -> array
```

6.10.6.1.1 segment dictionary

Segment with start, control-start, control-end, and end.

6.10.6.1.2 t int or float

Segment parameter in the range [0, 1].

6.10.7 cubic-tangent

Evaluate the tangent of a cubic segment at parameter t.

6.10.7.1 Parameters

```
cubic-tangent(  
  segment: dictionary ,  
  t: int | float  
) -> array
```

6.10.7.1.1 segment dictionary

Segment with start, control-start, control-end, and end.

6.10.7.1.2 t int or float

Segment parameter in the range [0, 1].

6.10.8 cubic-to

Build a cubic path element.

6.10.8.1 Parameters

```
cubic-to(  
  control-start: array ,  
  control-end: array ,  
  end: array  
) -> dictionary
```

6.10.8.1.1 control-start array

Cubic control point near the start point.

6.10.8.1.2 control-end array

Cubic control point near the endpoint.

6.10.8.1.3 end array

Cubic endpoint.

6.10.9 elements

Return the command elements that make up a Kurvst path.

6.10.9.1 Parameters

```
elements(path: dictionary) -> array
```

6.10.9.1.1 path dictionary

Kurvst path dictionary to inspect.

6.10.10 from-cubic

Build a path fragment from a cubic segment dictionary.

6.10.10.1 Parameters

```
from-cubic(segment: dictionary) -> dictionary
```

6.10.10.1.1 segment dictionary

Segment with start, control-start, control-end, and end.

6.10.11 from-elements

Build a path dictionary from an existing element array.

6.10.11.1 Parameters

```
from-elements(elements: array) -> dictionary
```

6.10.11.1.1 elements array

Array of Kurvst path elements.

6.10.12 hobby-spline

Construct a Hobby spline through an arbitrary point sequence.

6.10.12.1 Parameters

```
hobby-spline(  
  points: array,  
  omega: float,  
  accuracy: float  
) -> dictionary
```

6.10.12.1.1 points array

Two or more points for the open spline to pass through.

6.10.12.1.2 omega float

Hobby curl/tension parameter.

Default: 1.0

6.10.12.1.3 accuracy float

Geometry approximation accuracy passed to the Rust geometry engine.

Default: 0.001

6.10.13 hobby-through

Construct a cubic Hobby path through three points.

6.10.13.1 Parameters

```
hobby-through(  
  start: array ,  
  through: array ,  
  end: array ,  
  omega: float ,  
  accuracy: float  
) -> dictionary
```

6.10.13.1.1 start array

Start point.

6.10.13.1.2 through array

Intermediate point that the curve passes through.

6.10.13.1.3 end array

Endpoint.

6.10.13.1.4 omega float

Hobby curl/tension parameter.

Default: 1.0

6.10.13.1.5 accuracy float

Geometry approximation accuracy passed to the Rust geometry engine.

Default: 0.001

6.10.14 layer

Build a derived visible path layer.

6.10.14.1 Parameters

```
layer(  
  path: dictionary ,  
  offset: int float ,  
  length: none int float ,  
  ratio: none int float ,  
  resolve-length: string function ,  
  start-outset: int float ,  
  end-outset: int float ,  
  side-point: none array ,  
  accuracy: float ,  
  optimize: bool  
) -> dictionary
```

6.10.14.1.1 path dictionary

Base Kurvst path dictionary.

6.10.14.1.2 offset int or float

Signed normal offset distance.

Default: 0

6.10.14.1.3 length none or int or float

Fixed target visible length.

Default: none

6.10.14.1.4 ratio `none` or `int` or `float`

Relative target visible length as a fraction of the base length.

Default: `none`

6.10.14.1.5 resolve-length `string` or `function`

Resolution strategy for fixed and relative targets.

Default: `"min"`

6.10.14.1.6 start-outset `int` or `float`

Arc length removed from the start.

Default: `0`

6.10.14.1.7 end-outset `int` or `float`

Arc length removed from the end.

Default: `0`

6.10.14.1.8 side-point `none` or `array`

Optional point used to choose the sign of offset.

Default: `none`

6.10.14.1.9 accuracy `float`

Geometry approximation accuracy passed to the Rust geometry engine.

Default: `0.001`

6.10.14.1.10 optimize `bool`

Let Kurbo simplify/optimize fitted parallel paths.

Default: `true`

6.10.15 length

Compute the arc length of a path dictionary.

6.10.15.1 Parameters

```
length(  
  path: dictionary ,  
  accuracy: float  
) -> int float
```

6.10.15.1.1 path dictionary

Kurvst path dictionary to measure.

6.10.15.1.2 accuracy float

Arc-length approximation accuracy passed to the Rust geometry engine.

Default: 0.001

6.10.16 line

Build a straight-line path fragment.

6.10.16.1 Parameters

```
line(  
  start: array ,  
  end: array  
) -> dictionary
```

6.10.16.1.1 start array

Start point.

6.10.16.1.2 end array

Endpoint.

6.10.17 line-segment

Build a cubic segment dictionary for a straight line.

6.10.17.1 Parameters

```
line-segment(  
  start: array ,  
  end: array  
) -> dictionary
```

6.10.17.1.1 start array

Start point.

6.10.17.1.2 end array

Endpoint.

6.10.18 line-to

Build a line path element.

6.10.18.1 Parameters

```
line-to(end: array) -> dictionary
```

6.10.18.1.1 end array

Line endpoint.

6.10.19 move-to

Build a move path element.

6.10.19.1 Parameters

```
move-to(start: array) -> dictionary
```

6.10.19.1.1 start array

New current point and subpath start.

6.10.20 outset-point

Return from moved toward toward by distance.

6.10.20.1 Parameters

```
outset-point(  
  from: array ,  
  toward: array ,  
  distance: int float  
) -> array
```


6.10.20.1.1 from array

Point to move.

6.10.20.1.2 toward array

Target point that defines the direction.

6.10.20.1.3 distance int or float

Distance to move from from toward toward.

Default: 0

6.10.21 parallel

Generate a parallel path for a path.

6.10.21.1 Parameters

```
parallel(  
  path: dictionary ,  
  distance: int float ,  
  start-outset: int float ,  
  end-outset: int float ,  
  accuracy: float ,  
  optimize: bool  
) -> dictionary
```

6.10.21.1.1 path dictionary

Base Kurvst path dictionary.

6.10.21.1.2 distance int or float

Signed normal offset distance.

Default: 0

6.10.21.1.3 start-outset int or float

Arc length removed from the start before offsetting.

Default: 0

6.10.21.1.4 end-outset `int` or `float`

Arc length removed from the end before offsetting.

Default: `0`

6.10.21.1.5 accuracy `float`

Geometry approximation accuracy passed to the Rust geometry engine.

Default: `0.001`

6.10.21.1.6 optimize `bool`

Let Kurbo simplify/optimize the fitted path.

Default: `true`

6.10.22 path

Build a path dictionary from path fragments or elements.

6.10.22.1 Parameters

```
path(..parts: any) -> dictionary
```

6.10.22.1.1 ..parts `any`

Path fragments, path elements, or element arrays to concatenate.

6.10.23 pattern

Apply a repeated path pattern to a base path.

6.10.23.1 Parameters

```
pattern(  
  path: dictionary,  
  pattern: string dictionary,  
  amplitude: int float,  
  wavelength: int float,  
  phase: int float,  
  samples-per-period: int,  
  coil-longitudinal-scale: int float,  
  anchor-start: bool,  
  anchor-end: bool,  
  accuracy: float  
) -> dictionary
```

6.10.23.1.1 **path** dictionary

Base Kurvst path dictionary.

6.10.23.1.2 **pattern** string or dictionary

Pattern dictionary or built-in pattern name.

Default: "wave"

6.10.23.1.3 **amplitude** int or float

Normal amplitude of the pattern in path units.

Default: 0.1

6.10.23.1.4 **wavelength** int or float

Arc length of one pattern period.

Default: 1.0

6.10.23.1.5 **phase** int or float

Initial phase offset in pattern periods.

Default: 0

6.10.23.1.6 **samples-per-period** int

Samples per period for string-resolved smooth patterns.

Default: 16

6.10.23.1.7 **coil-longitudinal-scale** int or float

Longitudinal scale used when resolving the built-in coil pattern.

Default: 1.25

6.10.23.1.8 **anchor-start** bool

Force the generated path to start on the base path.

Default: true

6.10.23.1.9 anchor-end `bool`

Force the generated path to end on the base path.

Default: `true`

6.10.23.1.10 accuracy `float`

Geometry approximation accuracy passed to the Rust geometry engine.

Default: `0.001`

6.10.24 point

Build a numeric point tuple.

6.10.24.1 Parameters

```
point(  
  x: int float ,  
  y: int float  
) -> array
```

6.10.24.1.1 x `int` or `float`

X coordinate.

6.10.24.1.2 y `int` or `float`

Y coordinate.

6.10.25 points

Return the points visited by a Kurvst path's command stream.

6.10.25.1 Parameters

```
points(path: dictionary) -> array
```

6.10.25.1.1 path `dictionary`

Kurvst path dictionary to inspect.

6.10.26 quad

Build a quadratic path fragment.

6.10.26.1 Parameters

```
quad(  
  start: array ,  
  control: array ,  
  end: array  
) -> dictionary
```

6.10.26.1.1 start array

Start point.

6.10.26.1.2 control array

Quadratic control point.

6.10.26.1.3 end array

Endpoint.

6.10.27 quad-to

Build a quad path element.

6.10.27.1 Parameters

```
quad-to(  
  control: array ,  
  end: array  
) -> dictionary
```

6.10.27.1.1 control array

Quadratic control point.

6.10.27.1.2 end array

Quadratic endpoint.

6.10.28 resolve-length

Resolve a fixed and relative visible path length.

6.10.28.1 Parameters

```
resolve-length(  
    base-length: int float,  
    length: none int float,  
    ratio: none int float,  
    method: string function  
) -> none int float
```

6.10.28.1.1 base-length int or float

Full base path arc length.

6.10.28.1.2 length none or int or float

Fixed target arc length.

Default: none

6.10.28.1.3 ratio none or int or float

Relative target length as a fraction of base-length.

Default: none

6.10.28.1.4 method string or function

Resolution strategy for fixed and relative targets.

Default: "min"

6.10.29 segments

Return drawable cubic segments for any Kurvst path dictionary.

6.10.29.1 Parameters

```
segments(path: dictionary) -> array
```

6.10.29.1.1 path dictionary

Kurvst path dictionary to convert.

6.10.30 split-through

Split a path through a point sequence into per-span paths.

6.10.30.1 Parameters

```
split-through(  
  points: array ,  
  omega: float ,  
  start-outset: int float ,  
  end-outset: int float ,  
  accuracy: float  
) -> dictionary
```

6.10.30.1.1 points array

Two or more points for the curve to pass through.

6.10.30.1.2 omega float

Hobby curl/tension parameter.

Default: 1.0

6.10.30.1.3 start-outset int or float

Arc length removed from the first span start.

Default: 0

6.10.30.1.4 end-outset int or float

Arc length removed from the last span end.

Default: 0

6.10.30.1.5 accuracy float

Geometry approximation accuracy passed to the Rust geometry engine.

Default: 0.001

6.10.31 to-cetz

Draw a path dictionary through CeTZ.

6.10.31.1 Parameters

```
to-cetz(  
  path: dictionary ,  
  unit: int float length ratio ,  
  ..style: any  
) -> content
```

6.10.31.1.1 path dictionary

Kurvst path dictionary to draw.

6.10.31.1.2 unit int or float or length or ratio

Coordinate multiplier for emitted CeTZ points.

Default: 1

6.10.31.1.3 ..style any

CeTZ draw style arguments forwarded to merge-path.

6.10.32 to-cetz-data

Emit a Kurvst path as CeTZ path data.

6.10.32.1 Parameters

```
to-cetz-data(  
  path: dictionary ,  
  unit: int float length ratio  
) -> array
```

6.10.32.1.1 path dictionary

Kurvst path dictionary to convert.

6.10.32.1.2 unit int or float or length or ratio

Coordinate multiplier for emitted CeTZ points.

Default: 1

6.10.33 to-native

Emit a Kurvst path as native Typst curve content.

6.10.33.1 Parameters

```
to-native(  
  path: dictionary ,  
  unit: int float length ratio ,  
  ..style: any  
) -> content
```


6.10.33.1.1 path dictionary

Kurvst path dictionary to emit.

6.10.33.1.2 unit int or float or length or ratio

Coordinate multiplier for emitted Typst curve points.

Default: 1

6.10.33.1.3 ..style any

Native curve style arguments.

6.10.34 trim

Trim a path by arc length from each end.

6.10.34.1 Parameters

```
trim(  
  path: dictionary ,  
  start-outset: int float ,  
  end-outset: int float ,  
  accuracy: float  
) -> dictionary
```

6.10.34.1.1 path dictionary

Kurvst path dictionary to trim.

6.10.34.1.2 start-outset int or float

Arc length removed from the start.

Default: 0

6.10.34.1.3 end-outset int or float

Arc length removed from the end.

Default: 0

6.10.34.1.4 accuracy `float`

Arc-length approximation accuracy passed to the Rust geometry engine.

Default: `0.001`

6.10.35 wave

A smooth sinusoidal path pattern.

6.10.35.1 Parameters

`wave(samples-per-period: int) -> dictionary`

6.10.35.1.1 samples-per-period `int`

Number of samples used to approximate one wave period.

Default: `16`

6.10.36 zigzag

A straight-segment triangular path pattern.

6.10.36.1 Parameters

`zigzag() -> dictionary`

`dictionary`

6.10.37 layer-defaults

Default geometry options for derived path layers.

7 CLI, Rust, and Python APIs

7.1 Rust packages

The Rust API spans two primary packages.

- `gammaloops` contains the physics implementation, data model, integrands, integration, observables, and lower-level algorithms.
- `gammaloop-api` provides the supported state-loading API, session/command integration, CLI assembly, and the PyO3 extension used by the Python distribution.

Load a state with `StateLoadOption::load()`. Its options select a state directory, boot card, logging overrides, read-only behavior, and optional settings overrides. Once loaded, callers can create a CLI-style session or use dedicated structured operations. A raw command string is the compatibility fallback, not a replacement for a typed operation where one exists.

```

use std::path::PathBuf;
use gammaloop_api::{state::CommandHistory, StateLoadOption};

let mut loaded = StateLoadOption {
    state_folder: Some(PathBuf::from("./state")),
    ..StateLoadOption::default()
}.load()?;

let command = CommandHistory::from_raw_string("display settings global"?;
loaded.cli_session().execute_command(command)?;

```

Some Rust types are public so that GammaLoop’s packages can work together, but are not intended as stable integration points. Prefer the state-loading and structured-operation APIs described here; use the Rust orientation to choose a crate, then use that revision’s Rustdoc for exact signatures, trait implementations, and lower-level types.

7.1.1 Rust reference map

This authored map separates the supported workflow from the much larger compiled surface. Follow these exact Rustdoc paths instead of beginning in an arbitrary internal module:

- **Load and command a state:** begin with `StateLoadOption`, retain the returned `LoadedState`, and execute parsed `CommandHistory` values through `CliSession`.
- **Replay and inspect:** use `RunHistory` for run cards and `IntegrandInfo` for structured graph-group, orientation, loop-basis, cut, and threshold metadata.
- **Evaluate samples:** construct `EvaluateSamples` for the ordinary f64 boundary. Use `EvaluateSamplesPrecise` only when the caller must retain the active f64, f128, or arbitrary-precision result type.
- **Work below the facade:** the `gammaloop`rs Rustdoc includes the integrand contract, runtime settings, precision-specific result records, and the mergeable `histogram snapshot` boundary. Treat public items not named by a maintained workflow as implementation-facing unless their crate documents a stronger compatibility promise.

7.2 Python packaging

A standalone GammaLoop distribution

The Python distribution and import package are both named `gammaloop`. Its compiled backend is `gammaloop._gammaloop`, which applications normally access through the public package. This is separate from the `symbolica.community.*` modules used by `Spenso`, `Idenso`, and `Vakint`; installing `GammaLoop` does not install those community modules as independent packages.

The main Python entry point is `GammaLoopAPI`. Its constructor loads or creates one stateful session:

```

from gammaloop import GammaLoopAPI

gl = GammaLoopAPI(
    state_folder="./state",
    boot_commands_path="./run.toml",
)
gl.run("display settings global")

```

The Python package requires Python 3.11 or newer. Run just `build-api` when building the bindings from a source checkout.

7.2.1 Python reference map

The generated module reference covers every registered public export, but the objects form a few connected workflows rather than forty unrelated entry points:

- **Session lifecycle:** construct `GammaLoopAPI`, then use `run`, the history getters, and `SettingsValue` to automate the same state and commands as the CLI.
- **Point evaluation:** `evaluate_sample` returns `EvaluationResult`, whose `sample` is a `SampleEvaluationResult`. `evaluate_samples` returns `BatchEvaluationResult` with the same sample records and one batch-level observable snapshot.
- **Generated-integrand structure:** start at `IntegrandInfo`, then follow its graph groups into `graph`, `orientation`, `loop-momentum-basis`, `cut`, and `threshold` records. These objects describe compiled structure; they do not mutate it.
- **Events and observables:** evaluation records lead to `EventGroup` and `Event`. For caller-owned aggregation, `HistogramAccumulator` produces immutable `HistogramSnapshot` records with raw statistics that remain mergeable and reconstructible.

Full integrations still use the command interface

The current Python module does not expose a supported structured `integrate()` method. Run an integration through `GammaLoopAPI.run("integrate ...")` or the CLI and consume its persisted workspace/results. Several integration-result record classes are registered by the native module for ongoing API work, but without a public producer they are not part of the curated Python boundary yet.

Result ownership

Objects returned by evaluation and integration are snapshots. Mutating a Python list or dictionary derived from them does not update the live `GammaLoop` session. The explicit mutable exception is `HistogramAccumulator`, whose methods update caller-owned aggregation state.

7.3 Inspect and evaluate an existing state

The following workflow assumes that `./state` contains a generated integrand. Evaluation points are integrand-specific: their length must match the selected integrand and any discrete dimensions. Use the generated reference for the exact single-sample and batch contracts.

```
from collections.abc import Sequence

from gammaloop import GammaLoopAPI

api = GammaLoopAPI(state_folder="./state", read_only_state=True)
api.run("display processes")

print(api.get_run_history())
runtime = api.get_default_runtime_settings()
print(runtime.get("integrator.n_start"))

def evaluate_one(point: Sequence[float]) -> complex:
    result = api.evaluate_sample(
        point,
        process_id=0,
        minimal_output=True,
    )
    return result.integrand_result
```

Verification tier: compile

The documentation checks compile this example as Python syntax. They do not import or execute the native module: a runtime check additionally needs a built `GammaLoop` package, a provisioned backend and license, an existing generated state, and a point with the correct dimension.

The `run` method uses the same command language as the CLI and can change the in-memory state, settings, and run history. `read_only_state=True` protects files inside the active state directory; it does not make the Python object immutable or automatically persist the session. Inspect the live session through `get_run_history`, `get_global_settings`, and `get_active_command_blocks`.

For structured inspection, `get_integrand_info` describes the selected backend and graph structure, while `get_integrand_settings` and `get_default_runtime_settings` return detached, read-only `SettingsValue` snapshots. Use `get(path)`, attribute access, indexing, or `to_dict()` to read them; modifying derived Python values does not update the live session.

7.4 Sample evaluation and precision

`evaluate_sample` returns one sample result and the observable bundle for that one-sample batch. `evaluate_samples` accepts a two-dimensional NumPy array and returns per-sample results plus a batch-global observable bundle. Per-sample weights, discrete coordinates, graph names, and orientations must have the same row count as the input batch when provided. Graph and orientation selection applies only to momentum-space evaluation.

Both the ordinary Rust and Python endpoints expose numeric results through an `f64` contract, even when `use_arb_prec=True` requests arbitrary-precision internal evaluation. Evaluation may warm the integrand and update in-memory caches or observable snapshots, including in a read-only-state session. Rust-only callers that must retain the active precision use `evaluate_sample_precise` and `evaluate_samples_precise`.

For a complete, source-backed run card and scripts that expose event groups, cut metadata, selectors, and merged histogram snapshots, follow the [events and observables guide](#).

7.5 Symbolic conversion helpers

The supported module-level helpers are `atom_to_canonical_string` for parsing and canonicalizing a Symbolica expression, `evaluate_graph_overall_factor` for evaluating a graph's symbolic prefactor, and `to_dots` for rewriting tensor contractions into Idenso dot-product notation. Their generated entries document accepted strings, return values, failure modes, and task-oriented examples.

7.6 CLI and settings reference

7.6.1 Shell completion

`--completions` emits a static script from the same Clap command tree as `--help`. Load it for the current shell or save it in the shell's normal completion directory:

```
# Bash for the current session
source <(/gamma/loop --completions bash)

# Zsh for the current session
source <(/gamma/loop --completions zsh)
```

Fish, Elvish, PowerShell, and Nushell are also supported values. For Fish, pipe the output to `source`; for Nushell, save the generated module and then source it. Bash and Fish output also registers the repository wrapper spelling `./gamma/loop`.

Static and state-aware completion

The generated shell script covers executable subcommands, flags, enumerated values, and path hints. Completion inside the interactive GammaLoop prompt additionally knows the active state's processes, integrands, settings, model objects, graphs, cuts, and orientations. A static shell script cannot contain that changing session data.

Use `./gammaloop --help` and the generated [CLI and settings reference](#) for command names, aliases, flags, positional arguments, defaults, possible values, and setting paths. The tutorials explain how commands and settings combine into a persistent workflow. The [logging and diagnostics guide](#) covers startup precedence, selective tag filters, sink routing, release-build limitations, and copyable investigation patterns.

8 Version history

History coverage

GammaLoop does not yet have a consolidated changelog. Repository tags and their linked pull requests are the available release record. This page makes that gap explicit instead of presenting development documentation as a complete release history.

8.1 Components and version sources

- Repository revision: GammaLoop tags and releases.
- Python distribution: `pyproject.toml`.
- Core Rust library: `gammalooprs` package metadata.
- Rust/Python application API: `gammaloop-api` package metadata.

These components can evolve independently. Record all of them when reporting a result or asking for support.

8.2 Choosing the matching documentation

The `latest` channel follows current development. Documentation under `snapshots/<tag>` stays with a tagged release. Use the snapshot matching your checkout whenever command options, state files, or API signatures differ from `latest`.

8.3 Upgrade checklist

Release notes currently live with repository tags and their linked pull requests rather than in one consolidated changelog. Review them before upgrading, especially when you depend on persisted states, CLI scripts, Python bindings, or numerical results. Keep a copy of valuable state directories and re-run validation samples after changing versions.

8.4 Reproducibility record

Include the repository tag or commit, the Python distribution version, the `gammalooprs` version, the run card, and any relevant settings overrides. This makes a calculation easier to reproduce and distinguishes a software change from a configuration change.

Python API reference

The supported Python packages available in this version are listed below. The HTML manual provides a separate page for every public class and function; this printable edition keeps a compact inventory. Rust APIs use canonical Rustdoc in the HTML manual.

gammaloop-python

Exports registered by gammaloop.

AdditionalWeight

Named auxiliary complex weight attached to a generated event.

Named auxiliary complex weight attached to a generated event.

class AdditionalWeight:

Requires features: python_api, python_abi, python_stubgen

key

Kind: Getter

```
def key(self) -> str:
```

Stable weight name, optionally followed by colon-separated identifiers.

value

Kind: Getter

```
def value(self) -> Any:
```

Complex auxiliary event weight associated with ``key``.

Exhaustive reference · Source

BatchEvaluationResult

Per-sample evaluations paired with one observable snapshot for the complete batch.

Per-sample evaluations paired with one observable snapshot for the complete batch.

class BatchEvaluationResult:

Requires features: python_api, python_abi, python_stubgen

samples

Kind: Getter

```
def samples(self) -> list[SampleEvaluationResult]:
```

Evaluation records in the same order as the requested samples.

observables

Kind: Getter

```
def observables(self) -> dict:
```

Observable snapshots accumulated across the complete sample batch.

__str__

Kind: Method

```
def __str__(self) -> str:
```

Return a human-readable batch and observable summary.

__repr__

Kind: Method

```
def __repr__(self) -> str:
```

Return the human-readable batch and observable summary.

[Exhaustive reference](#) · [Source](#)

CutInfo

Process, graph, orientation, and loop-basis identifiers for one cut event.

Process, graph, orientation, and loop-basis identifiers for one cut event.

```
class CutInfo:
```

Requires features: python_api, python_abi, python_stubgen

incoming_pdgs

Kind: Getter

```
def incoming_pdgs(self) -> list[int]:
```

PDG identifiers aligned with ``Event.incoming\_momenta``.

outgoing_pdgs

Kind: Getter

```
def outgoing_pdgs(self) -> list[int]:
```

PDG identifiers aligned with ``Event.outgoing\_momenta``.

cut_id

Kind: Getter

```
def cut_id(self) -> int:
```

Zero-based cut identifier within the selected graph.

graph_id

Kind: Getter

```
def graph_id(self) -> int:
```

Zero-based graph identifier within the integrand.

graph_group_id

Kind: Getter

```
def graph_group_id(self) -> Optional[int]:
```

Graph-group identifier, when group metadata is available.

orientation_id

Kind: Getter

```
def orientation_id(self) -> Optional[int]:
```

Causal-flow orientation identifier, when sampled explicitly.

lmb_channel_id

Kind: Getter

```
def lmb_channel_id(self) -> Optional[int]:
```

Loop-momentum-basis multichannel identifier, when sampled explicitly.

lmb_channel_edge_ids

Kind: Getter

```
def lmb_channel_edge_ids(self) -> Optional[list[int]]:
```

Edge identifiers defining the selected loop-momentum basis, when available.

Exhaustive reference · Source

DotExportSettings

Controls which graph details and algebraic transformations are included in DOT exports.

Controls which graph details and algebraic transformations are included in DOT exports.

```
class DotExportSettings:
```

Requires features: python_api, python_abi, python_stubgen

combine_diagrams

Kind: Getter

```
def combine_diagrams(self) -> bool:
```

Write all diagrams of an integrand to one file during filesystem export.

combine_diagrams

Kind: Setter

```
def combine_diagrams(self, value: bool) -> None:
```

Write all diagrams of an integrand to one file during filesystem export.

value

Kind: Parameter

bool

with_uv

Kind: Getter

```
def with_uv(self) -> bool:
```

Request UV counterterm graph output; currently retained for compatibility.

with_uv

Kind: Setter

```
def with_uv(self, value: bool) -> None:
```

Request UV counterterm graph output; currently retained for compatibility.

value

Kind: Parameter

bool

output_full_numerator

Kind: Getter

```
def output_full_numerator(self) -> bool:
```

Add the complete symbolic numerator as the graph-level `full\_num` field.

output_full_numerator

Kind: Setter

```
def output_full_numerator(self, value: bool) -> None:
```

Add the complete symbolic numerator as the graph-level `full\_num` field.

value

Kind: Parameter

bool

split_xs_by_initial_states

Kind: Getter

```
def split_xs_by_initial_states(self) -> bool:
```

Split cross-section graphs by distinct initial-state assignments.

split_xs_by_initial_states

Kind: Setter

```
def split_xs_by_initial_states(self, value: bool) -> None:
```

Split cross-section graphs by distinct initial-state assignments.

value

Kind: Parameter

bool

do_gamma_algebra

Kind: Getter

```
def do_gamma_algebra(self) -> bool:
```

Simplify gamma-matrix algebra in the full numerator before formatting it.

do_gamma_algebra

Kind: Setter

```
def do_gamma_algebra(self, value: bool) -> None:
```

Simplify gamma-matrix algebra in the full numerator before formatting it.

value

Kind: Parameter

bool

do_color_algebra

Kind: Getter

```
def do_color_algebra(self) -> bool:
```

Simplify color algebra in the full numerator before formatting it.

do_color_algebra

Kind: Setter

```
def do_color_algebra(self, value: bool) -> None:
```

Simplify color algebra in the full numerator before formatting it.

value

Kind: Parameter

bool

include_autogenerated_fields

Kind: Getter

```
def include_autogenerated_fields(self) -> bool:
```

Include vertex and edge fields whose values GammaLoop generated automatically.

include_autogenerated_fields

Kind: Setter

```
def include_autogenerated_fields(self, value: bool) -> None:
```

Include vertex and edge fields whose values GammaLoop generated automatically.

value

Kind: Parameter

bool

__new__

Kind: AssociatedFunction

```
def __new__(cls) -> DotExportSettings:
```

Construct DOT export settings with GammaLoop's defaults.

Exhaustive reference · Source

EvaluationResult

Single-sample evaluation paired with the observable snapshot for that sample.

Single-sample evaluation paired with the observable snapshot for that sample.

class EvaluationResult:

Requires features: python_api, python_abi, python_stubgen

sample

Kind: Getter

```
def sample(self) -> SampleEvaluationResult:
```

Evaluation record for the requested sample.

integrand_result

Kind: Getter

```
def integrand_result(self) -> complex:
```

Complex integrand value before applying the parameterization Jacobian.

integrator_weight

Kind: Getter

```
def integrator_weight(self) -> float:
```

Monte Carlo weight supplied by the integrator, excluding the parameterization Jacobian.

generated_event_count**Kind:** Getter

```
def generated_event_count(self) -> Optional[int]:
```

Number of candidate events generated, or ``None`` when metadata was omitted.

accepted_event_count**Kind:** Getter

```
def accepted_event_count(self) -> Optional[int]:
```

Number of generated events accepted by selectors, or ``None`` without metadata.

event_processing_time_seconds**Kind:** Getter

```
def event_processing_time_seconds(self) -> Optional[float]:
```

Time spent building and selecting events, in seconds, or ``None`` without metadata.

parameterization_jacobian**Kind:** Getter

```
def parameterization_jacobian(self) -> Optional[float]:
```

Sample parameterization Jacobian, or ``None`` when no Jacobian was supplied.

is_nan**Kind:** Getter

```
def is_nan(self) -> Optional[bool]:
```

Whether evaluation metadata flagged a non-finite result, or ``None`` without metadata.

stability_results**Kind:** Getter

```
def stability_results(self) -> Optional[list[StabilityResult]]:
```

Per-precision stability attempts, or ``None`` when metadata was omitted.

event_groups**Kind:** Getter

```
def event_groups(self) -> list[EventGroup]:
```

Correlated event groups produced while evaluating this sample.

observables**Kind:** Getter

```
def observables(self) -> dict:
```

Observable snapshots produced from the same Monte Carlo sample.

__str__**Kind:** Method

```
def __str__(self) -> str:
```

Return a human-readable evaluation and observable summary.

__repr__

Kind: Method

```
def __repr__(self) -> str:
```

Return the human-readable evaluation and observable summary.

[Exhaustive reference](#) · [Source](#)

Event

Accepted cut event with momenta, its primary weight, and auxiliary weights.

Accepted cut event with momenta, its primary weight, and auxiliary weights.

```
class Event:
```

Requires features: python_api, python_abi, python_stubgen

incoming_momenta

Kind: Getter

```
def incoming_momenta(self) -> list[FourMomentum]:
```

Incoming four-momenta, aligned with ``cut\_info.incoming\_pdgs``.

outgoing_momenta

Kind: Getter

```
def outgoing_momenta(self) -> list[FourMomentum]:
```

Outgoing four-momenta, aligned with ``cut\_info.outgoing\_pdgs``.

cut_info

Kind: Getter

```
def cut_info(self) -> CutInfo:
```

Cut, graph, orientation, and loop-basis identifiers for this event.

weight

Kind: Getter

```
def weight(self) -> Any:
```

Primary complex event weight.

additional_weights

Kind: Getter

```
def additional_weights(self) -> list[AdditionalWeight]:
```

Named auxiliary complex weights such as threshold-counterterm contributions.

__str__

Kind: Method

```
def __str__(self) -> str:
```

Return a human-readable event summary.

__repr__

Kind: Method

```
def __repr__(self) -> str:
```

Return the human-readable event summary.

[Exhaustive reference](#) · [Source](#)

EventGroup

Correlated accepted events produced by one graph group for one sample.

Correlated accepted events produced by one graph group for one sample.

```
class EventGroup:
```

Requires features: python_api, python_abi, python_stubgen

events

Kind: Getter

```
def events(self) -> list[Event]:
```

Correlated accepted events in this group.

__str__

Kind: Method

```
def __str__(self) -> str:
```

Return a human-readable summary of the grouped events.

__repr__

Kind: Method

```
def __repr__(self) -> str:
```

Return the human-readable grouped-event summary.

[Exhaustive reference](#) · [Source](#)

FourMomentum

One energy-momentum four-vector returned with a generated event.

One energy-momentum four-vector returned with a generated event.

```
class FourMomentum:
```

Requires features: python_api, python_abi, python_stubgen

e

Kind: Getter

```
def e(self) -> float:
```

Energy component.

px

Kind: Getter

```
def px(self) -> float:
```

Momentum along the x axis.

py

Kind: Getter

```
def py(self) -> float:
```

Momentum along the y axis.

pz

Kind: Getter

```
def pz(self) -> float:
```

Momentum along the z axis.

Exhaustive reference · Source

GammaLoopAPI

Load, inspect, and evaluate one mutable GammaLoop session.

Load, inspect, and evaluate one mutable GammaLoop session.

One instance owns a GammaLoop state, run history, CLI settings, default runtime settings, and session state. Calls to ``run`` and the evaluation methods all act on that same in-memory session.

Notes

``read\_only\_state=True`` prevents writes inside the active state directory; it does not make this Python object immutable. Commands may still change in-memory settings, processes, or run history. Create separate instances when independent sessions are required.

```
class GammaLoopAPI:
```

Requires features: python_api, python_abi, python_stubgen

__new__

Kind: AssociatedFunction

```
def __new__(cls, state_folder: Optional[str | os.PathLike | pathlib.Path] = None,
boot_commands_path: Optional[str | os.PathLike | pathlib.Path] = None, model_file:
Optional[str | os.PathLike | pathlib.Path] = None, trace_logs_filename: Optional[str]
= None, level: Optional[LogLevel] = None, logfile_level: Optional[LogLevel] = None,
logging_prefix: object | None = None, read_only_state: bool = False,
settings_global_path: Optional[str | os.PathLike | pathlib.Path] = None,
settings_runtime_defaults_path: Optional[str | os.PathLike | pathlib.Path] = None,
clean_state: bool = False) -> GammaLoopAPI:
```

Load or create a GammaLoop state and initialize its CLI session.

Parameters

state_folder : path-like, optional

State directory to load or create. The default is ``./gammaloop\_state``.

boot_commands_path : path-like, optional

TOML run card whose commands are applied during startup.

model_file : path-like, optional

Model file override used while loading or initializing the state.

trace_logs_filename : str, optional

File receiving native trace records for this session.

level : LogLevel, optional

Terminal log-level override for this session.

logfile_level : LogLevel, optional

File log-level override for this session.

logging_prefix : object, optional

Native logging-prefix configuration.

read_only_state : bool, default=False

Prevent writes whose target lies inside the active state directory and disable file logging there. In-memory session changes remain possible.

`settings_global_path` : path-like, optional
TOML file overriding the global settings loaded at startup.

`settings_runtime_defaults_path` : path-like, optional
TOML file overriding the default runtime settings loaded at startup.

`clean_state` : bool, default=False
Remove the resolved state path before startup. This is destructive and cannot be combined with ``read\_only\_state=True``.

Returns

GammaLoopAPI

A stateful API instance sharing one state, history, and settings session.

Raises

Exception

If startup options conflict, the state or settings cannot be loaded, a boot command fails, or a boot card requests process exit.

Examples

Open an existing generated state without permitting writes to it::

```
api = GammaLoopAPI(state_folder="./state", read_only_state=True)
```

state_folder

Kind: Parameter

Optional[str | os.PathLike | pathlib.Path]

Default: None

State directory to load or create. The default is ``./gammaloop\_state``.

boot_commands_path

Kind: Parameter

Optional[str | os.PathLike | pathlib.Path]

Default: None

TOML run card whose commands are applied during startup.

model_file

Kind: Parameter

Optional[str | os.PathLike | pathlib.Path]

Default: None

Model file override used while loading or initializing the state.

trace_logs_filename

Kind: Parameter

Optional[str]

Default: None

File receiving native trace records for this session.

level

Kind: Parameter

Optional[LogLevel]

Default: None

Terminal log-level override for this session.

logfile_level

Kind: Parameter

Optional[LogLevel]

Default: None

File log-level override for this session.

logging_prefix

Kind: Parameter

object | None

Default: None

Native logging-prefix configuration.

read_only_state

Kind: Parameter

bool

Default: False

Prevent writes whose target lies inside the active state directory and disable file logging there. In-memory session changes remain possible.

settings_global_path

Kind: Parameter

Optional[str | os.PathLike | pathlib.Path]

Default: None

TOML file overriding the global settings loaded at startup.

settings_runtime_defaults_path

Kind: Parameter

Optional[str | os.PathLike | pathlib.Path]

Default: None

TOML file overriding the default runtime settings loaded at startup.

clean_state

Kind: Parameter

bool

Default: False

Remove the resolved state path before startup. This is destructive and cannot be combined with ``read\_only\_state=True``.

evaluate_sample

Kind: Method

```
def evaluate_sample(self, point: Sequence[float], process_id: Optional[int] = None,
integrand_name: Optional[str] = None, use_arb_prec: bool = False, minimal_output:
bool = False, return_events: Optional[bool] = None, momentum_space: bool = False,
integrator_weight: Optional[float] = None, discrete_dim: Optional[Sequence[int]] =
None, graph_name: Optional[str] = None, orientation: Optional[int] = None) ->
EvaluationResult:
```

Evaluate one integration or momentum-space sample.

Parameters

point : Sequence[float]

Coordinates for one sample. In integration space, the length must match the selected integrand and ``discrete\_dim``. In momentum space, values are grouped as ``(px, py, pz)`` and the length must be a multiple of 3.

process_id : int, optional

Process containing the integrand. Supply this when selection is ambiguous.

integrand_name : str, optional

Integrand to evaluate. Supply this when selection is ambiguous.

use_arb_prec : bool, default=False

Request arbitrary-precision internal evaluation.

minimal_output : bool, default=False

Omit the optional evaluation metadata from the returned sample.

return_events : bool, optional

Temporarily override event generation for this call. The integrand setting is restored afterward.

momentum_space : bool, default=False

Interpret ``point`` as flattened three-momenta instead of integration-space coordinates.

integrator_weight : float, optional

Weight associated with this sample. The default is 1.0.

discrete_dim : Sequence[int], optional

Discrete integration coordinates used to determine the expected dimension.

graph_name : str, optional

Graph selected for momentum-space evaluation.

orientation : int, optional

Orientation index for ``graph\_name`` in momentum-space evaluation.

Returns

EvaluationResult

The sample evaluation and the observable snapshot for its one-row batch.

Raises

Exception

If integrand selection is ambiguous or invalid, dimensions do not match, graph or orientation selection is invalid, or warm-up/evaluation fails.

Notes

Arbitrary precision affects the internal calculation, but Python-visible numeric fields use the package's ``float64`` output contract. Evaluation may warm the integrand and update in-memory caches or observable snapshots even in a read-only-state session.

Examples

Evaluate a point whose dimension matches integrand 0::

```
result = api.evaluate_sample(point, process_id=0, minimal_output=True)
value = result.integrand_result
```

point

Kind: Parameter

Sequence[float]

Coordinates for one sample. In integration space, the length must match the selected integrand and ``discrete\_dim``. In momentum space, values are grouped as ``(px, py, pz)`` and the length must be a multiple of 3.

process_id

Kind: Parameter

Optional[int]

Default: None

Process containing the integrand. Supply this when selection is ambiguous.

integrand_name

Kind: Parameter

Optional[str]

Default: None

Integrand to evaluate. Supply this when selection is ambiguous.

use_arb_prec

Kind: Parameter

bool

Default: False

Request arbitrary-precision internal evaluation.

minimal_output

Kind: Parameter

bool

Default: False

Omit the optional evaluation metadata from the returned sample.

return_events

Kind: Parameter

Optional[bool]

Default: None

Temporarily override event generation for this call. The integrand setting is restored afterward.

momentum_space

Kind: Parameter

bool

Default: False

Interpret ``point`` as flattened three-momenta instead of integration-space coordinates.

integrator_weight

Kind: Parameter

Optional[float]

Default: None

Weight associated with this sample. The default is 1.0.

discrete_dim

Kind: Parameter

Optional[Sequence[int]]

Default: None

Discrete integration coordinates used to determine the expected dimension.

graph_name

Kind: Parameter

Optional[str]

Default: None

Graph selected for momentum-space evaluation.

orientation

Kind: Parameter

Optional[int]

Default: None

Orientation index for ``graph\_name`` in momentum-space evaluation.

evaluate_samples

Kind: Method

```
def evaluate_samples(self, points: numpy.ndarray[numpy.float64], process_id:
Optional[int] = None, integrand_name: Optional[str] = None, use_arb_prec: bool =
False, minimal_output: bool = False, return_events: Optional[bool] = None,
momentum_space: bool = False, integrator_weights:
Optional[numpy.ndarray[numpy.float64]] = None, discrete_dims:
numpy.ndarray[numpy.unsignedinteger] | None = None, graph_names:
Optional[Sequence[Optional[str]]] = None, orientations:
Optional[Sequence[Optional[int]]] = None) -> BatchEvaluationResult:
```

Evaluate a batch of integration or momentum-space samples.

Parameters

`points` : `numpy.ndarray[numpy.float64]`

Two-dimensional array with one sample per row. Integration-space columns must match the selected integrand; momentum-space columns are flattened

```(px, py, pz)``` groups.

`process_id` : `int`, optional

Process containing the integrand. Supply this when selection is ambiguous.

`integrand_name` : `str`, optional

Integrand to evaluate. Supply this when selection is ambiguous.

`use_arb_prec` : `bool`, default=False

Request arbitrary-precision internal evaluation.

`minimal_output` : `bool`, default=False

Omit the optional evaluation metadata from every returned sample.

`return_events` : `bool`, optional

Temporarily override event generation for this call. The integrand setting is restored afterward.

`momentum_space` : `bool`, default=False

Interpret each row as flattened three-momenta instead of integration-space coordinates.

`integrator_weights` : `numpy.ndarray[numpy.float64]`, optional

One weight per row. Defaults to 1.0 for every sample.

`discrete_dims` : `numpy.ndarray[numpy.unsignedinteger]`, optional

Two-dimensional array with one row of discrete coordinates per sample.

`graph_names` : `Sequence[str | None]`, optional

Momentum-space graph selection for each sample.

`orientations` : `Sequence[int | None]`, optional

Momentum-space orientation selection for each sample.

#### Returns

-----

`BatchEvaluationResult`

Per-sample evaluations and one observable snapshot for the complete batch.

#### Raises

-----

`Exception`

If integrand selection is ambiguous or invalid, array or option lengths do not match, dimensions are invalid, or warm-up/evaluation fails.

#### Notes

-----

Arbitrary precision affects the internal calculation, but Python-visible numeric fields use the package's ```float64``` output contract. Evaluation may update in-memory caches or observable snapshots in a read-only-state session.

#### Examples

-----

Evaluate a caller-provided two-dimensional array::

```
points = numpy.asarray(points, dtype=numpy.float64)
result = api.evaluate_samples(points, process_id=0, minimal_output=True)
```

**points****Kind:** Parameter

numpy.ndarray[numpy.float64]

Two-dimensional array with one sample per row. Integration-space columns must match the selected integrand; momentum-space columns are flattened ``(px, py, pz)`` groups.

**process\_id****Kind:** Parameter

Optional[int]

**Default:** None

Process containing the integrand. Supply this when selection is ambiguous.

**integrand\_name****Kind:** Parameter

Optional[str]

**Default:** None

Integrand to evaluate. Supply this when selection is ambiguous.

**use\_arb\_prec****Kind:** Parameter

bool

**Default:** False

Request arbitrary-precision internal evaluation.

**minimal\_output****Kind:** Parameter

bool

**Default:** False

Omit the optional evaluation metadata from every returned sample.

**return\_events****Kind:** Parameter

Optional[bool]

**Default:** None

Temporarily override event generation for this call. The integrand setting is restored afterward.

**momentum\_space****Kind:** Parameter

bool

**Default:** False

Interpret each row as flattened three-momenta instead of integration-space coordinates.

### **integrator\_weights**

**Kind:** Parameter

Optional[numpy.ndarray[numpy.float64]]

**Default:** None

One weight per row. Defaults to 1.0 for every sample.

### **discrete\_dims**

**Kind:** Parameter

numpy.ndarray[numpy.unsignedinteger] | None

**Default:** None

Two-dimensional array with one row of discrete coordinates per sample.

### **graph\_names**

**Kind:** Parameter

Optional[Sequence[Optional[str]]]

**Default:** None

Momentum-space graph selection for each sample.

### **orientations**

**Kind:** Parameter

Optional[Sequence[Optional[int]]]

**Default:** None

Momentum-space orientation selection for each sample.

### **import\_graphs**

**Kind:** Method

```
def import_graphs(self, graphs: str, process_name: Optional[str] = None, process_id:
Optional[int] = None, integrand_name: Optional[str] = None, format: str = 'dot',
overwrite: bool = False, append: bool = False) -> None:
```

Import DOT graphs into a new or existing process/integrand collection.

Parameters

-----

graphs : str

DOT file path when ``format="dot"`` or inline DOT text when  
``format="string"``.

process\_name, process\_id : str or int, optional

Process to create or update. Inline text requires one of these selectors.

integrand\_name : str, optional

Integrand within the selected process.

format : {"dot", "string"}, default="dot"

Select file-backed or inline input.

overwrite, append : bool, default=False

Replace an existing collection or append to it; these modes conflict.

Raises

-----

Exception

If selectors conflict, the source is missing or malformed, or importing fails.

### **graphs**

**Kind:** Parameter

str

DOT file path when ``format="dot"`` or inline DOT text when ``format="string"``.

### **process\_name**

**Kind:** Parameter

Optional[str]

**Default:** None

Process to create or update. Inline text requires one of these selectors.

### **process\_id**

**Kind:** Parameter

Optional[int]

**Default:** None

Process to create or update. Inline text requires one of these selectors.

### **integrand\_name**

**Kind:** Parameter

Optional[str]

**Default:** None

Integrand within the selected process.

### **format**

**Kind:** Parameter

str

**Default:** 'dot'

Select file-backed or inline input.

### **overwrite**

**Kind:** Parameter

bool

**Default:** False

Replace an existing collection or append to it; these modes conflict.

### **append**

**Kind:** Parameter

bool

**Default:** False

Replace an existing collection or append to it; these modes conflict.



### **get\_lmbs**

**Kind:** Method

```
def get_lmbs(self, graphs: str, format: str = 'dot') -> list[list[tuple[list[int], list[int], dict[int, tuple[list[int], list[int]]]]]]:
```

Generate loop-momentum bases for each supplied graph.

Parameters

-----

graphs : str

DOT file path or inline DOT text, as selected by ``format``.

format : {"dot", "string"}, default="dot"

Select file-backed or inline input.

Returns

-----

list

Per graph, a list of ``(loop\_edges, external\_edges, edge\_signatures)`` tuples. Each signature contains its loop and external coefficients.

### **graphs**

**Kind:** Parameter

str

DOT file path or inline DOT text, as selected by ``format``.

### **format**

**Kind:** Parameter

str

**Default:** 'dot'

Select file-backed or inline input.

### **get\_orientations**

**Kind:** Method

```
def get_orientations(self, graph_name: str, process_id: Optional[int] = None, integrand_name: Optional[str] = None) -> list[dict[int, int]]:
```

Return the causal-flow orientations generated for one graph.

Each returned dictionary maps an edge id to ``1`` (default), ``-1`` (reversed), or ``0`` (undirected). Supply process and integrand selectors when the active state does not identify a unique integrand.

Parameters

-----

graph\_name : str

Name of the graph within the selected integrand.

process\_id : int, optional

Numeric process identifier; omit when process selection is unambiguous.

integrand\_name : str, optional

Integrand containing the graph; omit when integrand selection is unambiguous.

Returns

-----

list[dict[int, int]]

One edge-direction mapping per generated orientation.

### **graph\_name**

**Kind:** Parameter

str

Name of the graph within the selected integrand.

### **process\_id**

**Kind:** Parameter

Optional[int]

**Default:** None

Numeric process identifier; omit when process selection is unambiguous.

### **integrand\_name**

**Kind:** Parameter

Optional[str]

**Default:** None

Integrand containing the graph; omit when integrand selection is unambiguous.

### **get\_model**

**Kind:** Method

def get\_model(self) -> str:

Serialize the active physics model as JSON.

Returns

-----

str

JSON representation of the model currently owned by this session.

### **evaluate**

**Kind:** Method

def evaluate(self, process\_id: Optional[int] = None, graphs\_group\_name: Optional[str] = None, result\_path: Optional[str | os.PathLike | pathlib.Path] = None, numerical: bool = True, number\_of\_terms\_in\_epsilon\_expansion: Optional[int] = None) -> str:

Evaluate one generated graph group as a symbolic or numerical expression.

Parameters

-----

process\_id : int, optional

Process containing the requested graph group.

graphs\_group\_name : str, optional

Group to evaluate when the current state is ambiguous.

result\_path : path-like, optional

Optional destination for the evaluated expression.

numerical : bool, default=True

Evaluate numerically instead of retaining a symbolic result.

number\_of\_terms\_in\_epsilon\_expansion : int, optional

Truncate the dimensional-regulator expansion to this many terms.

Returns

-----

str

Canonical Symbolica representation of the result.

**process\_id**

**Kind:** Parameter

Optional[int]

**Default:** None

Process containing the requested graph group.

**graphs\_group\_name**

**Kind:** Parameter

Optional[str]

**Default:** None

Group to evaluate when the current state is ambiguous.

**result\_path**

**Kind:** Parameter

Optional[str | os.PathLike | pathlib.Path]

**Default:** None

Optional destination for the evaluated expression.

**numerical**

**Kind:** Parameter

bool

**Default:** True

Evaluate numerically instead of retaining a symbolic result.

**number\_of\_terms\_in\_epsilon\_expansion**

**Kind:** Parameter

Optional[int]

**Default:** None

Truncate the dimensional-regulator expansion to this many terms.

**import\_model**

**Kind:** Method

```
def import_model(self, model_specifier: str | os.PathLike | pathlib.Path,
simplify_model: bool = True) -> None:
```

Replace the active model from a GammaLoop model file or supported model source.

``simplify\_model`` applies the standard symbolic simplification pass while importing. Existing process data may no longer be compatible with a replaced model, so import the model before generating processes.

## Parameters

-----

**model\_specifier** : path-like

Path or model specifier accepted by GammaLoop's model importer.

**simplify\_model** : bool, default=True

Apply the standard symbolic simplification pass while importing.

## **model\_specifier**

**Kind:** Parameter

str | os.PathLike | pathlib.Path

Path or model specifier accepted by GammaLoop's model importer.

## **simplify\_model**

**Kind:** Parameter

bool

**Default:** True

Apply the standard symbolic simplification pass while importing.

## **list\_outputs**

**Kind:** Method

def list\_outputs(self) -> tuple[dict[str, int], dict[str, int]]:

List generated amplitude and cross-section names with their process ids.

## Returns

-----

tuple[dict[str, int], dict[str, int]]

Amplitude mapping followed by the cross-section mapping.

## **get\_integrand\_info**

**Kind:** Method

def get\_integrand\_info(self, process\_id: Optional[int] = None, integrand\_name:

Optional[str] = None) -> IntegrandInfo:

Describe the selected generated integrand and its graph structure.

## Parameters

-----

**process\_id** : int, optional

Process containing the integrand. Supply this when selection is ambiguous.

**integrand\_name** : str, optional

Integrand to inspect. Supply this when selection is ambiguous.

## Returns

-----

IntegrandInfo

Structured process, backend, graph, orientation, cut, and size metadata.

## Raises

-----

Exception

If no unique generated integrand matches the selection.

## Examples

-----

Inspect the active evaluation backend and graph count::

```
info = api.get_integrand_info(process_id=0)
print(info.active_f64_backend, info.graph_count)
```

### **process\_id**

**Kind:** Parameter

Optional[int]

**Default:** None

Process containing the integrand. Supply this when selection is ambiguous.

### **integrand\_name**

**Kind:** Parameter

Optional[str]

**Default:** None

Integrand to inspect. Supply this when selection is ambiguous.

### **get\_integrand\_settings**

**Kind:** Method

```
def get_integrand_settings(self, process_id: Optional[int] = None, integrand_name:
Optional[str] = None) -> SettingsValue:
```

Return a detached, read-only snapshot of one integrand's settings.

### Parameters

-----

process\_id : int, optional

Process containing the integrand. Supply this when selection is ambiguous.

integrand\_name : str, optional

Integrand whose settings are required.

### Returns

-----

SettingsValue

Serialized settings snapshot supporting ``get(path)``, attribute access, indexing, and ``to\_dict()``. Mutating it does not update the live session.

### Raises

-----

Exception

If selection fails, the integrand has not been generated, or its settings cannot be serialized.

## Examples

-----

Read a nested setting without changing the live integrand::

```
settings = api.get_integrand_settings(process_id=0)
print(settings.get("general.generate_events"))
```

**process\_id****Kind:** Parameter

Optional[int]

**Default:** None

Process containing the integrand. Supply this when selection is ambiguous.

**integrand\_name****Kind:** Parameter

Optional[str]

**Default:** None

Integrand whose settings are required.

**get\_run\_history****Kind:** Method

```
def get_run_history(self) -> str:
```

Render the current in-memory run history as TOML.

Returns

-----

str

TOML representation of the history owned by this API instance.

Raises

-----

Exception

If the current history cannot be serialized.

Notes

-----

This reports the live session, including commands run through ``run``; it does not imply that the history has been persisted to the state directory.

Examples

-----

Capture a reproducible run card from the current session::

```
run_card_toml = api.get_run_history()
```

**get\_global\_settings****Kind:** Method

```
def get_global_settings(self) -> str:
```

Render the current effective CLI and global settings as TOML.

Returns

-----

str

TOML representation of the settings used by this API session.

Raises

-----

#### Exception

If the settings cannot be rendered.

#### Examples

-----

Record the effective settings after applying startup overrides::

```
settings_toml = api.get_global_settings()
```

#### **get\_active\_command\_blocks**

**Kind:** Method

```
def get_active_command_blocks(self) -> dict[str, list[str]]:
```

Return the named command blocks in the current run history.

#### Returns

-----

```
dict[str, list[str]]
```

Mapping from block name to its rendered CLI commands.

#### Examples

-----

Inspect the commands currently grouped under each block::

```
for name, commands in api.get_active_command_blocks().items():
 print(name, commands)
```

#### **get\_default\_runtime\_settings**

**Kind:** Method

```
def get_default_runtime_settings(self) -> SettingsValue:
```

Return a detached, read-only snapshot of the default runtime settings.

#### Returns

-----

```
SettingsValue
```

Serialized settings including defaults. Use ``get(path)`` or ``to\_dict()`` to inspect it; changes to derived Python values do not affect the session.

#### Raises

-----

#### Exception

If the settings cannot be serialized.

#### Examples

-----

Inspect a runtime default by its documented settings path::

```
runtime = api.get_default_runtime_settings()
print(runtime.get("integrator.n_start"))
```

#### **get\_dot\_files**

**Kind:** Method

```
def get_dot_files(self, process: Optional[int | str] = None, integrand_name:
Optional[str] = None, settings: DotExportSettings = ...) -> str:
```

Render a selected amplitude or cross section as Graphviz DOT text.

#### Parameters

-----

process : int or str, optional

Process id or name; omit only when selection is unambiguous.

integrand\_name : str, optional

Integrand to render.

settings : DotExportSettings, optional

Controls diagram combination, UV terms, algebra, and generated fields.

#### Returns

-----

str

DOT source suitable for Graphviz or GammaLoop's drawing pipeline.

#### **process**

**Kind:** Parameter

Optional[int | str]

**Default:** None

Process id or name; omit only when selection is unambiguous.

#### **integrand\_name**

**Kind:** Parameter

Optional[str]

**Default:** None

Integrand to render.

#### **settings**

**Kind:** Parameter

DotExportSettings

**Default:** ...

Controls diagram combination, UV terms, algebra, and generated fields.

#### **run**

**Kind:** Method

def run(self, command: str) -> None:

Parse and execute a CLI command in this API instance's session.

#### Parameters

-----

command : str

GammaLoop CLI command text.

#### Returns

-----

None

The command's normal output is emitted through the configured CLI/logging sinks; inspect structured state through the corresponding getter methods.



Raises

-----

Exception

If the command cannot be parsed or execution fails.

Notes

-----

Commands share and may mutate the instance's in-memory state, settings, and run history. ``read\_only\_state=True`` only blocks writes inside the active state directory. The API does not automatically persist the session after this call; persistence depends on the executed command's explicit output behavior.

Examples

-----

Display the processes loaded in the current state::

```
api.run("display processes")
```

**command**

**Kind:** Parameter

str

GammaLoop CLI command text.

**generate\_cff**

**Kind:** Method

```
def generate_cff(self, dot_string: str, subgraph_nodes: Sequence[str],
reverse_dangling: Sequence[int], orientation_pattern: Optional[str] = None) ->
list[tuple[dict[int, int], str]]:
```

Build a causal-flow expression from an inline DOT graph or one of its subgraphs.

Parameters

-----

**dot\_string** : str

Inline DOT graph using particles from the active model.

**subgraph\_nodes** : Sequence[str]

Vertex names retained in the subgraph; an empty sequence selects all nodes.

**reverse\_dangling** : Sequence[int]

Dangling edge ids whose orientation is reversed.

**orientation\_pattern** : str, optional

Pattern restricting returned causal-flow orientations.

Returns

-----

```
list[tuple[dict[int, int], str]]
```

Edge-direction maps paired with their energy-denominator expressions.

**dot\_string**

**Kind:** Parameter

str

Inline DOT graph using particles from the active model.

**subgraph\_nodes**

**Kind:** Parameter

Sequence[str]

Vertex names retained in the subgraph; an empty sequence selects all nodes.

**reverse\_dangling**

**Kind:** Parameter

Sequence[int]

Dangling edge ids whose orientation is reversed.

**orientation\_pattern**

**Kind:** Parameter

Optional[str]

**Default:** None

Pattern restricting returned causal-flow orientations.

**generate\_cff\_as\_json\_string**

**Kind:** Method

```
def generate_cff_as_json_string(self, dot_string: str, subgraph_nodes: Sequence[str],
reverse_dangling: Sequence[int], orientation_pattern: Optional[str] = None) -> str:
```

Serialize a causal-flow expression and its surfaces as JSON.

This accepts the same graph, subgraph, and dangling-edge inputs as ``generate\_cff``. The current JSON representation is intended for GammaLoop tooling and may contain internal structural details.

Parameters

-----

**dot\_string** : str

Inline DOT graph using particles from the active model.

**subgraph\_nodes** : Sequence[str]

Vertex names retained in the subgraph; an empty sequence selects all nodes.

**reverse\_dangling** : Sequence[int]

Dangling edge ids whose orientation is reversed.

**orientation\_pattern** : str, optional

Pattern restricting returned causal-flow orientations.

Returns

-----

str

JSON representation of the causal-flow expression and E-surfaces.

**dot\_string**

**Kind:** Parameter

str

Inline DOT graph using particles from the active model.

**subgraph\_nodes**

**Kind:** Parameter

Sequence[str]

Vertex names retained in the subgraph; an empty sequence selects all nodes.

## **reverse\_dangling**

**Kind:** Parameter

Sequence[int]

Dangling edge ids whose orientation is reversed.

## **orientation\_pattern**

**Kind:** Parameter

Optional[str]

**Default:** None

Pattern restricting returned causal-flow orientations.

Exhaustive reference · Source

## **HistogramAccumulator**

Mutable continuous or discrete histogram accumulator with sample-level statistics.

Mutable continuous or discrete histogram accumulator with sample-level statistics.

class HistogramAccumulator:

**Requires features:** python\_api, python\_abi, python\_stubgen

### **continuous**

**Kind:** AssociatedFunction

```
def continuous(title: str, x_min: float, x_max: float, n_bins: int, type_description: str = 'AL', phase: str = 'real', value_transform: str = 'identity', log_x_axis: bool = False, log_y_axis: bool = True) -> HistogramAccumulator:
```

Create an evenly binned continuous histogram accumulator.

Parameters

-----

title : str

Human-readable title used by snapshots and exported output.

x\_min : float

Lower edge of the histogram range.

x\_max : float

Upper edge of the histogram range.

n\_bins : int

Number of equal-width in-range bins.

type\_description : str, default="AL"

HwU ``TYPE@`` metadata.

phase : {"real", "imag"}, default="real"

Component of each complex event weight to accumulate.

value\_transform : {"identity", "log10"}, default="identity"

Transformation applied to coordinates before binning.

log\_x\_axis : bool, default=False

Request logarithmic horizontal-axis rendering.

log\_y\_axis : bool, default=True

Request logarithmic vertical-axis rendering.

### **title**

**Kind:** Parameter

str

Human-readable title used by snapshots and exported output.

**x\_min**

**Kind:** Parameter

float

Lower edge of the histogram range.

**x\_max**

**Kind:** Parameter

float

Upper edge of the histogram range.

**n\_bins**

**Kind:** Parameter

int

Number of equal-width in-range bins.

**type\_description**

**Kind:** Parameter

str

**Default:** 'AL'

HwU ``TYPE@`` metadata.

**phase**

**Kind:** Parameter

str

**Default:** 'real'

Component of each complex event weight to accumulate.

**value\_transform**

**Kind:** Parameter

str

**Default:** 'identity'

Transformation applied to coordinates before binning.

**log\_x\_axis**

**Kind:** Parameter

bool

**Default:** False

Request logarithmic horizontal-axis rendering.

**log\_y\_axis**

**Kind:** Parameter

bool

**Default:** True

Request logarithmic vertical-axis rendering.

**discrete**

**Kind:** AssociatedFunction

```
def discrete(title: str, min_bin_id: int, max_bin_id: int, ordering: str =
'ascending_bin_id', labels: Optional[Sequence[str]] = None, type_description: str =
'AL', phase: str = 'real', log_y_axis: bool = True) -> HistogramAccumulator:
```

Create a discrete histogram accumulator over an inclusive integer range.

Parameters

-----

**title** : str

Human-readable title used by snapshots and exported output.

**min\_bin\_id** : int

Smallest accepted bin identifier.

**max\_bin\_id** : int

Largest accepted bin identifier, inclusive.

**ordering** : {"ascending\_bin\_id", "value\_descending", "abs\_value\_descending"},

default="ascending\_bin\_id"

Ordering used when bins are returned or exported.

**labels** : Sequence[str], optional

One label per bin, ordered by ascending bin identifier.

**type\_description** : str, default="AL"

HwU ``TYPE@`` metadata.

**phase** : {"real", "imag"}, default="real"

Component of each complex event weight to accumulate.

**log\_y\_axis** : bool, default=True

Request logarithmic vertical-axis rendering.

Raises

-----

**ValueError**

If the range, label count, ordering, or phase is invalid.

**title**

**Kind:** Parameter

str

Human-readable title used by snapshots and exported output.

**min\_bin\_id**

**Kind:** Parameter

int

Smallest accepted bin identifier.

**max\_bin\_id**

**Kind:** Parameter

int

Largest accepted bin identifier, inclusive.

**ordering****Kind:** Parameter

str

**Default:** 'ascending\_bin\_id'

Ordering used when bins are returned or exported.

**labels****Kind:** Parameter

Optional[Sequence[str]]

**Default:** None

One label per bin, ordered by ascending bin identifier.

**type\_description****Kind:** Parameter

str

**Default:** 'AL'

HwU ``TYPE@`` metadata.

**phase****Kind:** Parameter

str

**Default:** 'real'

Component of each complex event weight to accumulate.

**log\_y\_axis****Kind:** Parameter

bool

**Default:** True

Request logarithmic vertical-axis rendering.

**snapshot****Kind:** Method

```
def snapshot(self) -> HistogramSnapshot:
```

Return an immutable snapshot including committed and pending samples.

**merge\_in\_place****Kind:** Method

```
def merge_in_place(self, other: HistogramAccumulator) -> None:
```

Move pending samples from another compatible accumulator into this one.

Parameters

-----

other : HistogramAccumulator

Compatible accumulator whose pending samples are consumed.

Raises

-----

ValueError

If histogram definitions differ.

**other**

**Kind:** Parameter

HistogramAccumulator

Compatible accumulator whose pending samples are consumed.

**rebin**

**Kind:** Method

def rebin(self, contiguous\_bins: int) -> HistogramAccumulator:

Return a copy with each run of adjacent continuous bins combined.

Discrete histograms are copied unchanged.

Parameters

-----

contiguous\_bins : int

Positive number of old bins per new bin; it must divide the bin count.

**contiguous\_bins**

**Kind:** Parameter

int

Positive number of old bins per new bin; it must divide the bin count.

**rescale**

**Kind:** Method

def rescale(self, factor: float) -> None:

Multiply all accumulated weights by a constant in place.

Parameters

-----

factor : float

Scale applied to weight sums; squared-weight sums use its square.

**factor**

**Kind:** Parameter

float

Scale applied to weight sums; squared-weight sums use its square.

**change\_bin\_ordering**

**Kind:** Method

def change\_bin\_ordering(self, ordering: str) -> None:

Change the display ordering of a discrete histogram.

Parameters

-----

ordering : {"ascending\_bin\_id", "value\_descending", "abs\_value\_descending"}

New ordering for snapshots and exported output.

Raises

-----

ValueError

If this is a continuous histogram or the name is invalid.

### **ordering**

**Kind:** Parameter

str

New ordering for snapshots and exported output.

### **update\_results**

**Kind:** Method

def update\_results(self) -> None:

Commit pending samples to the accumulator's completed-result counters.

### **fill\_continuous\_sample**

**Kind:** Method

def fill\_continuous\_sample(self, entries: Sequence[tuple[float, float]]) -> None:

Add one independent continuous sample containing coordinate-weight pairs.

Parameters

-----

entries : Sequence[tuple[float, float]]

``(coordinate, projected\_weight)`` entries for this sample.

Raises

-----

ValueError

If this is not a continuous histogram.

### **entries**

**Kind:** Parameter

Sequence[tuple[float, float]]

``(coordinate, projected\_weight)`` entries for this sample.

### **fill\_discrete\_sample**

**Kind:** Method

def fill\_discrete\_sample(self, entries: Sequence[tuple[int, float]]) -> None:

Add one independent discrete sample containing bin-weight pairs.

Parameters

-----

entries : Sequence[tuple[int, float]]

``(bin\_id, projected\_weight)`` entries for this sample.

Raises

-----

ValueError

If this is not a discrete histogram.



**entries**

**Kind:** Parameter

Sequence[tuple[int, float]]

`((bin_id, projected_weight))` entries for this sample.

Exhaustive reference · Source

**HistogramBin**

Raw statistics and optional coordinate metadata for one histogram bin.

Raw statistics and optional coordinate metadata for one histogram bin.

class HistogramBin:

**Requires features:** python\_api, python\_abi, python\_stubgen

**x\_min**

**Kind:** Getter

def x\_min(self) -> Optional[float]:

Lower coordinate edge, or `None` for a discrete or overflow bin.

**x\_max**

**Kind:** Getter

def x\_max(self) -> Optional[float]:

Upper coordinate edge, or `None` for a discrete or underflow bin.

**bin\_id**

**Kind:** Getter

def bin\_id(self) -> Optional[int]:

Discrete bin identifier, or `None` for a continuous bin.

**label**

**Kind:** Getter

def label(self) -> Optional[str]:

Optional label of a discrete bin.

**entry\_count**

**Kind:** Getter

def entry\_count(self) -> int:

Number of event entries accumulated in this bin.

**sum\_weights**

**Kind:** Getter

def sum\_weights(self) -> float:

Sum of per-sample projected weights for this bin.

**sum\_weights\_squared**

**Kind:** Getter

def sum\_weights\_squared(self) -> float:

Sum of squared per-sample projected weights for this bin.

**mitigated\_fill\_count****Kind:** Getter

```
def mitigated_fill_count(self) -> int:
```

Number of fills produced by paired boundary-misbinning mitigation.

**average****Kind:** Method

```
def average(self, sample_count: int) -> float:
```

Compute this bin's Monte Carlo mean from its raw weight sum.

Parameters

-----

sample\_count : int

Number of statistically independent samples represented by the histogram.

Returns

-----

float

`sum_weights / sample_count`, or zero when `sample_count` is zero.

**sample\_count****Kind:** Parameter

int

Number of statistically independent samples represented by the histogram.

**error****Kind:** Method

```
def error(self, sample_count: int) -> float:
```

Compute the standard error of this bin's Monte Carlo mean.

Parameters

-----

sample\_count : int

Number of statistically independent samples represented by the histogram.

Returns

-----

float

Sample-mean standard error, or zero when it cannot be estimated reliably.

**sample\_count****Kind:** Parameter

int

Number of statistically independent samples represented by the histogram.

Exhaustive reference · Source

**HistogramSnapshot**

Immutable, mergeable histogram snapshot containing raw per-bin statistics.

Immutable, mergeable histogram snapshot containing raw per-bin statistics.

```
class HistogramSnapshot:
```

**Requires features:** python\_api, python\_abi, python\_stubgen

**kind**

**Kind:** Getter

```
def kind(self) -> str:
```

```
 ``"continuous"`` or ``"discrete"``.
```

**title**

**Kind:** Getter

```
def title(self) -> str:
```

```
 Human-readable histogram title.
```

**type\_description**

**Kind:** Getter

```
def type_description(self) -> str:
```

```
 HwU ``TYPE@`` description retained with the histogram.
```

**phase**

**Kind:** Getter

```
def phase(self) -> str:
```

```
 Complex-weight component accumulated: ``"real"`` or ``"imag"``.
```

**value\_transform**

**Kind:** Getter

```
def value_transform(self) -> str:
```

```
 Coordinate transform: ``"identity"`` or ``"log10"``.
```

**supports\_misbinning\_mitigation**

**Kind:** Getter

```
def supports_misbinning_mitigation(self) -> bool:
```

```
 Whether paired fills can mitigate bin-boundary instability.
```

**x\_min**

**Kind:** Getter

```
def x_min(self) -> Optional[float]:
```

```
 Lower bound of a continuous histogram, otherwise ``None``.
```

**x\_max**

**Kind:** Getter

```
def x_max(self) -> Optional[float]:
```

```
 Upper bound of a continuous histogram, otherwise ``None``.
```

**sample\_count**

**Kind:** Getter

```
def sample_count(self) -> int:
```

Number of statistically independent Monte Carlo samples represented.

### **log\_x\_axis**

**Kind:** Getter

```
def log_x_axis(self) -> bool:
```

Whether compatible renderers should use a logarithmic horizontal axis.

### **log\_y\_axis**

**Kind:** Getter

```
def log_y_axis(self) -> bool:
```

Whether compatible renderers should use a logarithmic vertical axis.

### **discrete\_min\_bin\_id**

**Kind:** Getter

```
def discrete_min_bin_id(self) -> Optional[int]:
```

Smallest discrete bin identifier, otherwise ``None``.

### **discrete\_ordering**

**Kind:** Getter

```
def discrete_ordering(self) -> Optional[str]:
```

Discrete display ordering, otherwise ``None``.

### **bins**

**Kind:** Getter

```
def bins(self) -> list[HistogramBin]:
```

Raw statistics for every in-range bin, in display order.

### **underflow\_bin**

**Kind:** Getter

```
def underflow_bin(self) -> HistogramBin:
```

Raw statistics for entries below a continuous histogram's range.

### **overflow\_bin**

**Kind:** Getter

```
def overflow_bin(self) -> HistogramBin:
```

Raw statistics for entries above a continuous histogram's range.

### **statistics**

**Kind:** Getter

```
def statistics(self) -> HistogramStats:
```

Histogram-wide entry and mitigation counters.

Exhaustive reference · Source

### **HistogramStats**

Aggregate entry, NaN, and misbinning-mitigation counts for a histogram.

Aggregate entry, NaN, and misbinning-mitigation counts for a histogram.

```
class HistogramStats:
```

**Requires features:** python\_api, python\_abi, python\_stubgen

**in\_range\_entry\_count**

**Kind:** Getter

```
def in_range_entry_count(self) -> int:
```

Number of event entries assigned to in-range bins.

**nan\_value\_count**

**Kind:** Getter

```
def nan_value_count(self) -> int:
```

Number of entries skipped because their coordinate was non-finite.

**mitigated\_pair\_count**

**Kind:** Getter

```
def mitigated_pair_count(self) -> int:
```

Number of entry pairs processed by boundary-misbinning mitigation.

Exhaustive reference · Source

**IntegrandActiveThresholdCut**

Viable cut associated with a threshold E-surface.

Viable cut associated with a threshold E-surface.

```
class IntegrandActiveThresholdCut:
```

**Requires features:** python\_api, python\_abi, python\_stubgen

**cut\_id**

**Kind:** Getter

```
def cut_id(self) -> int:
```

Zero-based identifier of a cut on which the E-surface remains viable.

**can\_become\_pinched**

**Kind:** Getter

```
def can_become_pinched(self) -> bool:
```

Whether this cut can make the E-surface pinched for some kinematics.

Exhaustive reference · Source

**IntegrandCut**

Cut edges, raising power, and threshold associations for one cross-section cut.

Cut edges, raising power, and threshold associations for one cross-section cut.

```
class IntegrandCut:
```

**Requires features:** python\_api, python\_abi, python\_stubgen

**cut\_id**

**Kind:** Getter

```
def cut_id(self) -> int:
```

Zero-based cut identifier within the representative graph.

#### **edge\_ids**

**Kind:** Getter

```
def edge_ids(self) -> list[int]:
```

Edge identifiers crossed by the cut.

#### **raising\_power**

**Kind:** Getter

```
def raising_power(self) -> int:
```

Maximum multiplicity of the cut's related threshold group.

#### **left\_thresholds**

**Kind:** Getter

```
def left_thresholds(self) -> list[IntegrandCutThreshold]:
```

Threshold associations on the left side of the cut.

#### **right\_thresholds**

**Kind:** Getter

```
def right_thresholds(self) -> list[IntegrandCutThreshold]:
```

Threshold associations on the right side of the cut.

Exhaustive reference · Source

#### **IntegrandCutThreshold**

Classification and boundary edges of one threshold associated with a cut.

Classification and boundary edges of one threshold associated with a cut.

```
class IntegrandCutThreshold:
```

**Requires features:** python\_api, python\_abi, python\_stubgen

#### **esurface\_id**

**Kind:** Getter

```
def esurface_id(self) -> int:
```

Threshold E-surface identifier within the graph group.

#### **status**

**Kind:** Getter

```
def status(self) -> str:
```

Viability classification for this threshold-cut association.

#### **cut\_boundary\_edge\_ids**

**Kind:** Getter

```
def cut_boundary_edge_ids(self) -> list[int]:
```

Cut edges that bound the associated threshold region.

#### **threshold\_boundary\_edge\_ids**

**Kind:** Getter

```
def threshold_boundary_edge_ids(self) -> list[int]:
```

Threshold edges that bound the associated threshold region.

#### **invariant\_bound\_is\_applicable**

**Kind:** Getter

```
def invariant_bound_is_applicable(self) -> bool:
```

Whether the invariant-bound criterion applies to this association.

Exhaustive reference · Source

#### **IntegrandGraph**

Identity and master-graph status of one graph in an integrand.

Identity and master-graph status of one graph in an integrand.

```
class IntegrandGraph:
```

**Requires features:** python\_api, python\_abi, python\_stubgen

#### **graph\_id**

**Kind:** Getter

```
def graph_id(self) -> int:
```

Zero-based graph identifier within the integrand.

#### **name**

**Kind:** Getter

```
def name(self) -> str:
```

Persistent graph name.

#### **is\_master**

**Kind:** Getter

```
def is_master(self) -> bool:
```

Whether this graph is the representative graph of its group.

Exhaustive reference · Source

#### **IntegrandGraphGroup**

Graphs and shared orientation, loop-basis, threshold, and cut data in one group.

Graphs and shared orientation, loop-basis, threshold, and cut data in one group.

```
class IntegrandGraphGroup:
```

**Requires features:** python\_api, python\_abi, python\_stubgen

#### **group\_id**

**Kind:** Getter

```
def group_id(self) -> int:
```

Zero-based graph-group identifier within the integrand.

#### **graphs**

**Kind:** Getter

```
def graphs(self) -> list[IntegrandGraph]:
```

Graphs in this group, with the representative graph marked as master.

#### **orientation\_edge\_ids**

**Kind:** Getter

```
def orientation_edge_ids(self) -> list[int]:
```

Edge identifiers that establish the ordering of every orientation signature.

#### **orientations**

**Kind:** Getter

```
def orientations(self) -> list[IntegrandOrientation]:
```

Available causal-flow orientations for the representative graph.

#### **loop\_momentum\_bases**

**Kind:** Getter

```
def loop_momentum_bases(self) -> list[IntegrandLoopMomentumBasis]:
```

Available loop-momentum bases for the representative graph.

#### **threshold\_esurface\_ids**

**Kind:** Getter

```
def threshold_esurface_ids(self) -> list[int]:
```

Ordered identifiers of threshold E-surfaces retained for the group.

#### **threshold\_esurfaces**

**Kind:** Getter

```
def threshold_esurfaces(self) -> list[IntegrandThresholdEsurface]:
```

Structured metadata for the retained threshold E-surfaces.

#### **cuts**

**Kind:** Getter

```
def cuts(self) -> list[IntegrandCut]:
```

Cross-section cuts; empty for amplitude graph groups.

Exhaustive reference · Source

#### **IntegrandInfo**

Structured description of a generated integrand and its evaluation backend.

Structured description of a generated integrand and its evaluation backend.

```
class IntegrandInfo:
```

**Requires features:** python\_api, python\_abi, python\_stubgen

#### **process\_id**

**Kind:** Getter

```
def process_id(self) -> int:
```

Numeric identifier of the process that owns the integrand.

#### **process\_name**

**Kind:** Getter



```
def process_name(self) -> str:
```

Persistent folder name of the owning process.

**integrand\_name**

**Kind:** Getter

```
def integrand_name(self) -> str:
```

Canonical name of the resolved integrand.

**kind**

**Kind:** Getter

```
def kind(self) -> str:
```

``"amplitude"`` or ``"cross section"``.

**generation\_backend**

**Kind:** Getter

```
def generation_backend(self) -> str:
```

Compilation backend frozen into the generated integrand.

**generation\_compile\_options**

**Kind:** Getter

```
def generation_compile_options(self) -> Optional[str]:
```

Backend-specific compilation options, when configured.

**active\_f64\_backend**

**Kind:** Getter

```
def active_f64_backend(self) -> str:
```

Floating-point evaluator backend currently active for f64 evaluation.

**graph\_count**

**Kind:** Getter

```
def graph_count(self) -> int:
```

Number of amplitude graphs or cross-section supergraphs.

**graph\_group\_count**

**Kind:** Getter

```
def graph_group_count(self) -> int:
```

Number of graph groups in the generated integrand.

**record\_size\_bytes**

**Kind:** Getter

```
def record_size_bytes(self) -> int:
```

Serialized integrand-record size in bytes, excluding its compiled evaluator.

**graph\_groups**

**Kind:** Getter

```
def graph_groups(self) -> list[IntegrandGraphGroup]:
```

Per-group graph, orientation, basis, threshold, and cut metadata.

[Exhaustive reference](#) · [Source](#)

### **IntegrandLoopMomentumBasis**

Loop-momentum basis and optional multichannel identifier for a graph group.

Loop-momentum basis and optional multichannel identifier for a graph group.

```
class IntegrandLoopMomentumBasis:
```

**Requires features:** `python_api`, `python_abi`, `python_stubgen`

#### **basis\_id**

**Kind:** Getter

```
def basis_id(self) -> int:
```

Zero-based loop-momentum-basis identifier on the representative graph.

#### **channel\_id**

**Kind:** Getter

```
def channel_id(self) -> Optional[int]:
```

Effective multichannel identifier, or ``None`` when this basis is not sampled.

#### **edge\_ids**

**Kind:** Getter

```
def edge_ids(self) -> list[int]:
```

Edge identifiers whose momenta form the basis.

#### **matches\_generation\_basis**

**Kind:** Getter

```
def matches_generation_basis(self) -> bool:
```

Whether this is the loop-momentum basis used during integrand generation.

[Exhaustive reference](#) · [Source](#)

### **IntegrandOrientation**

Edge-direction signature for one causal-flow orientation.

Edge-direction signature for one causal-flow orientation.

```
class IntegrandOrientation:
```

**Requires features:** `python_api`, `python_abi`, `python_stubgen`

#### **orientation\_id**

**Kind:** Getter

```
def orientation_id(self) -> int:
```

Zero-based orientation identifier within the graph group.

#### **signature**

**Kind:** Getter

```
def signature(self) -> list[int]:
```

Direction per ``IntegrandGraphGroup.orientation\_edge\_ids``: ``1``, ``-1``, or ``0``.

[Exhaustive reference](#) · [Source](#)

### **IntegrandThresholdEsurface**

Threshold E-surface metadata and the cuts on which it remains active.

Threshold E-surface metadata and the cuts on which it remains active.

```
class IntegrandThresholdEsurface:
```

**Requires features:** python\_api, python\_abi, python\_stubgen

#### **esurface\_id**

**Kind:** Getter

```
def esurface_id(self) -> int:
```

E-surface identifier within the graph group.

#### **representative\_graph\_id**

**Kind:** Getter

```
def representative_graph_id(self) -> int:
```

Graph whose local E-surface supplies ``edge\_ids``.

#### **edge\_ids**

**Kind:** Getter

```
def edge_ids(self) -> list[int]:
```

Edge identifiers contributing energies to this E-surface.

#### **classification**

**Kind:** Getter

```
def classification(self) -> Optional[str]:
```

Kinematic existence class, or ``None`` when no static classification was computed.

#### **active\_cuts**

**Kind:** Getter

```
def active_cuts(self) -> list[IntegrandActiveThresholdCut]:
```

Cuts on which this E-surface remains viable.

[Exhaustive reference](#) · [Source](#)

### **SampleEvaluationResult**

Numerical value, metadata, stability records, and generated events for one sample.

Numerical value, metadata, stability records, and generated events for one sample.

```
class SampleEvaluationResult:
```

**Requires features:** python\_api, python\_abi, python\_stubgen

#### **integrand\_result**

**Kind:** Getter

```
def integrand_result(self) -> complex:
```

Complex integrand value before applying the parameterization Jacobian.

**integrator\_weight****Kind:** Getter

```
def integrator_weight(self) -> float:
```

Monte Carlo weight supplied by the integrator, excluding the parameterization Jacobian.

**generated\_event\_count****Kind:** Getter

```
def generated_event_count(self) -> Optional[int]:
```

Number of candidate events generated, or ``None`` when metadata was omitted.

**accepted\_event\_count****Kind:** Getter

```
def accepted_event_count(self) -> Optional[int]:
```

Number of generated events accepted by selectors, or ``None`` without metadata.

**event\_processing\_time\_seconds****Kind:** Getter

```
def event_processing_time_seconds(self) -> Optional[float]:
```

Time spent building and selecting events, in seconds, or ``None`` without metadata.

**parameterization\_jacobian****Kind:** Getter

```
def parameterization_jacobian(self) -> Optional[float]:
```

Sample parameterization Jacobian, or ``None`` when no Jacobian was supplied.

**is\_nan****Kind:** Getter

```
def is_nan(self) -> Optional[bool]:
```

Whether evaluation metadata flagged a non-finite result, or ``None`` without metadata.

**stability\_results****Kind:** Getter

```
def stability_results(self) -> Optional[list[StabilityResult]]:
```

Per-precision stability attempts, or ``None`` when metadata was omitted.

**event\_groups****Kind:** Getter

```
def event_groups(self) -> list[EventGroup]:
```

Correlated event groups produced while evaluating this sample.

**\_\_str\_\_****Kind:** Method

```
def __str__(self) -> str:
```

Return a human-readable evaluation summary.

**\_\_repr\_\_**

**Kind:** Method

```
def __repr__(self) -> str:
```

Return the human-readable evaluation summary.

Exhaustive reference · Source

**SettingsValue**

Read-only, detached view over serialized GammaLoop settings.

Read-only, detached view over serialized GammaLoop settings.

```
class SettingsValue:
```

**Requires features:** python\_api, python\_abi, python\_stubgen

**path**

**Kind:** Getter

```
def path(self) -> str:
```

Dot-separated location of this value within the settings tree.

**kind**

**Kind:** Getter

```
def kind(self) -> str:
```

JSON value kind: ``object``, ``array``, ``string``, ``number``, ``boolean``, or ``null``.

**is\_object**

**Kind:** Method

```
def is_object(self) -> bool:
```

Return whether this value is an object with named children.

**is\_array**

**Kind:** Method

```
def is_array(self) -> bool:
```

Return whether this value is an array with indexed children.

**is\_scalar**

**Kind:** Method

```
def is_scalar(self) -> bool:
```

Return whether this value is a scalar JSON value, including ``null``.

**keys**

**Kind:** Method

```
def keys(self) -> list[str]:
```

Return this object's child keys in lexical order.

Raises

-----

**TypeError**

If this value is not an object.

**get****Kind:** Method

```
def get(self, path: str) -> Any:
```

Resolve a dot-separated descendant path.

An empty path returns this value. Objects become ``SettingsValue`` views, while scalar values become their ordinary Python equivalents.

Parameters

-----

path : str

Relative path such as ``"sampling.parameterization"``.

Raises

-----

KeyError

If the path does not exist below this value.

**path****Kind:** Parameter

str

Relative path such as ``"sampling.parameterization"``.

**to\_dict****Kind:** Method

```
def to_dict(self) -> Any:
```

Convert the complete detached value tree to ordinary Python containers and scalars.

**\_\_getattr\_\_****Kind:** Method

```
def __getattr__(self, name: str) -> Any:
```

Resolve an object child by attribute name.

Parameters

-----

name : str

Immediate child key to resolve.

**name****Kind:** Parameter

str

Immediate child key to resolve.

**\_\_dir\_\_****Kind:** Method

```
def __dir__(self) -> list[str]:
```

Return object-child names in lexical order, or an empty list for other kinds.

**\_\_len\_\_****Kind:** Method

```
def __len__(self) -> int:
```

Return the number of object or array children.

**`__str__`**

**Kind:** Method

```
def __str__(self) -> str:
```

Return the detached settings value as pretty-printed JSON.

**`__repr__`**

**Kind:** Method

```
def __repr__(self) -> str:
```

Return a compact representation containing this value's path and kind.

**`__getitem__`**

**Kind:** Method

```
def __getitem__(self, key: Any, /) -> Any:
```

Resolve a relative string path or array index.

Parameters

-----

**key** : str or int

Descendant path for an object, or zero-based index for an array.

**key**

**Kind:** Parameter

Any

Descendant path for an object, or zero-based index for an array.

Exhaustive reference · Source

**StabilityResult**

Outcome and cost of one numerical-stability precision level.

Outcome and cost of one numerical-stability precision level.

class StabilityResult:

**Requires features:** python\_api, python\_abi, python\_stubgen

**precision**

**Kind:** Getter

```
def precision(self) -> str:
```

Numerical precision used for this stability level.

**estimated\_relative\_accuracy**

**Kind:** Getter

```
def estimated_relative_accuracy(self) -> Optional[float]:
```

Estimated relative accuracy, or ``None`` when it could not be estimated.

**status**

**Kind:** Getter

```
def status(self) -> str:
```

Human-readable stability outcome and number of rotated samples.

#### **sample\_count**

**Kind:** Getter

```
def sample_count(self) -> int:
```

Number of rotated samples evaluated at this precision level.

#### **total\_time\_seconds**

**Kind:** Getter

```
def total_time_seconds(self) -> float:
```

Total wall-clock time spent at this precision level, in seconds.

[Exhaustive reference](#) · [Source](#)

#### **atom\_to\_canonical\_string**

Parse a Symbolica expression and return its canonical string form.

Parse a Symbolica expression and return its canonical string form.

##### Parameters

-----

**atom\_str** : str

Symbolica expression to parse and normalize.

##### Returns

-----

str

The parsed expression in canonical string form.

##### Raises

-----

##### Exception

If ``atom\_str`` is not a valid Symbolica expression.

##### Examples

-----

Normalize a symbolic expression before comparing or storing it::

```
 atom_to_canonical_string("x + x")
```

```
def atom_to_canonical_string(atom_str: str) -> str:
```

**Requires features:** `python_api`, `python_abi`, `python_stubgen`

[Exhaustive reference](#) · [Source](#)

#### **evaluate\_graph\_overall\_factor**

Evaluate a graph's symbolic overall factor and return its canonical form.

Evaluate a graph's symbolic overall factor and return its canonical form.

##### Parameters

-----

**overall\_factor** : str

Symbolica expression used as the graph's overall factor.



Returns

-----

str

The evaluated factor in canonical Symbolica form.

Raises

-----

Exception

If the expression cannot be parsed or evaluated.

Examples

-----

Evaluate the symmetry and fermion-loop factors attached to a graph::

```
evaluate_graph_overall_factor(
 "AutG(2)^-1*InternalFermionLoopSign(-1)"
)
```

```
def evaluate_graph_overall_factor(overall_factor: str) -> str:
```

**Requires features:** python\_api, python\_abi, python\_stubgen

Exhaustive reference · Source

**to\_dots**

Rewrite a Symbolica tensor expression into Idenso dot-product notation.

Rewrite a Symbolica tensor expression into Idenso dot-product notation.

Parameters

-----

atom\_str : str

Symbolica tensor expression whose repeated indices encode contractions.

Returns

-----

str

The equivalent expression using Idenso dot-product notation.

Raises

-----

Exception

If ``atom\_str`` cannot be parsed or converted.

Examples

-----

Rewrite a contraction before passing it to an Idenso workflow::

```
to_dots("p(mu) * q(mu)")
```

```
def to_dots(atom_str: str) -> str:
```

**Requires features:** python\_api, python\_abi, python\_stubgen

Exhaustive reference · Source

## CLI commands and settings

Commands and settings available in this version are listed below.

### Command tree

Command	Description
gammaLoop	An implementation of the Local Unitarity method for computing differential collider cross-sections.
gammaLoop display	Inspect models, processes, integrands, settings, and named session data
gammaLoop display model	Show model particles, parameters, couplings, and vertices for the selected context
gammaLoop display processes	List processes currently stored in the active state
gammaLoop display integrand	Show generated integrands, graph groups, resources, and selected detail tables
gammaLoop display quantities	List configured named quantities, or inspect one quantity in a selected process
gammaLoop display observables	List configured observables, or inspect one observable in a selected process
gammaLoop display selectors	List configured event selectors, or inspect one selector in a selected process
gammaLoop display command_block	Show one stored command block, or list every block when no name is supplied
gammaLoop display settings	Show effective global, default-runtime, or process-specific settings
gammaLoop display settings global	Show effective global generation, logging, and parallelization settings
gammaLoop display settings default-runtime	Show the default runtime settings inherited by newly generated integrands
gammaLoop display settings process	Show effective settings for one generated process or integrand
gammaLoop duplicate	Copy generated data to a new process or integrand name
gammaLoop duplicate integrand	Copy one generated integrand and its process context under new names
gammaLoop set	Change global, runtime, model, or process settings
gammaLoop set base-dir	Set the base directory for gammaloop state and logs
gammaLoop set global	Change global generation, logging, and parallelism settings
gammaLoop set global file	Merge a TOML file into the current global or default-runtime settings
gammaLoop set global string	Load settings from TOML content passed as a CLI string
gammaLoop set global kv	Merge one or more dotted `KEY=VALUE` assignments into the current settings

Command	Description
gammaLoop set global defaults	Sync process runtime settings from the current default runtime settings
gammaLoop set global stored	Reload the corresponding settings from the active state's stored file
gammaLoop set default-runtime	Change the runtime settings inherited by newly generated integrands
gammaLoop set default-runtime file	Merge a TOML file into the current global or default-runtime settings
gammaLoop set default-runtime string	Load settings from TOML content passed as a CLI string
gammaLoop set default-runtime kv	Merge one or more dotted `KEY=VALUE` assignments into the current settings
gammaLoop set default-runtime defaults	Sync process runtime settings from the current default runtime settings
gammaLoop set default-runtime stored	Reload the corresponding settings from the active state's stored file
gammaLoop set model	Assign model parameters in the selected context or reset them to defaults
gammaLoop set process	Change runtime settings or named analysis objects for one process or integrand
gammaLoop set process file	Merge a TOML file into the selected process or integrand settings
gammaLoop set process string	Load settings from TOML content passed as a CLI string
gammaLoop set process kv	Merge one or more dotted `KEY=VALUE` assignments into the selected settings
gammaLoop set process defaults	Sync process runtime settings from the current default runtime settings
gammaLoop set process stored	Reload the selected process or integrand settings from its stored file
gammaLoop set process add	Add a named quantity/observable/selector
gammaLoop set process add quantity	Add a named derived quantity to the selected process settings
gammaLoop set process add observable	Add a named observable to the selected process settings
gammaLoop set process add selector	Add a named event selector to the selected process settings
gammaLoop set process update	Update a named quantity/observable/selector
gammaLoop set process update quantity	Update fields of an existing named quantity
gammaLoop set process update observable	Update fields of an existing named observable
gammaLoop set process update selector	Update fields of an existing named selector
gammaLoop set process remove	Remove a named quantity/observable/selector

Command	Description
<code>gammaLoop set process remove quantity</code>	Remove a named quantity from the selected process settings
<code>gammaLoop set process remove observable</code>	Remove a named observable from the selected process settings
<code>gammaLoop set process remove selector</code>	Remove a named selector from the selected process settings
<code>gammaLoop import</code>	Import models or previously generated graphs into the active state
<code>gammaLoop import model</code>	Load a UFO model and make it the active model for subsequent generation
<code>gammaLoop import graphs</code>	Load serialized graph data into a new or existing process
<code>gammaLoop save</code>	Export graph data, standalone evaluators, schemas, or the current state
<code>gammaLoop save dot</code>	Export generated graphs as DOT files and drawing templates
<code>gammaLoop save uv-forest</code>	Export the ultraviolet forest for selected generated graphs
<code>gammaLoop save standalone</code>	Export an integrand as a standalone evaluator or serialized data file
<code>gammaLoop save state</code>	Persist the active state, settings, and replayable run history
<code>gammaLoop save schema</code>	Regenerate the JSON schema files for CLI and runtime configuration
<code>gammaLoop run</code>	Execute a stored command block or an inline command list
<code>gammaLoop start_commands_block</code>	Begin recording subsequent commands under a reusable block name
<code>gammaLoop finish_commands_block</code>	Stop recording commands and store the active command block
<code>gammaLoop remove</code>	Remove generated processes from the active state
<code>gammaLoop remove processes</code>	Remove one integrand, one process, or all processes selected by the supplied target
<code>gammaLoop integrate</code>	Integrate one or more generated process integrands
<code>gammaLoop generate</code>	Generate cross-section or amplitude integrands from a process specification
<code>gammaLoop generate xs</code>	Generate a cross-section integrand through the forward-scattering construction
<code>gammaLoop generate amp</code>	Generate an amplitude integrand for the supplied process specification
<code>gammaLoop generate existing</code>	Create another integrand from an existing process without regenerating its graphs
<code>gammaLoop select</code>	Select graph groups and write the selection to a new process or integrand

Command	Description
gammaLoop quit	End the interactive GammaLoop session without executing further commands
gammaLoop inspect	Inspect integrand contributions at one phase-space point or momentum configuration
gammaLoop approach	Sample an integrand while approaching a phase-space point along selected scaling axes
gammaLoop evaluate	Evaluate a vacuum amplitude analytically or numerically with Vakint
gammaLoop renormalize	Compute and export ultraviolet renormalization contributions for an amplitude
gammaLoop bench	Measure raw integrand evaluation throughput over random samples and selected worker counts
gammaLoop profile	Profile ultraviolet or infrared scaling limits of a generated integrand
gammaLoop profile ultra-violet	Sample ultraviolet scaling rays and report the fitted large-momentum behavior
gammaLoop profile bulk	Probe multiple infrared scaling limits and report their fitted behavior in one run
gammaLoop batch	Evaluate a process-definition file against an HPC batch input and write named results
gammaLoop !	Run an operating-system shell command from the interactive GammaLoop session

## Settings

Path	Type	Default
cli.global.display_directive	string	"info"
cli.global.generation.compile.compilation_mode	enum	"symjit"
cli.global.generation.compile.compiler	string	"g++"
cli.global.generation.compile.custom_compiler	array<string>	[]
cli.global.generation.compile.fast_build	boolean	true
cli.global.generation.compile.compiler_optimization_level	enum	"O2"
cli.global.generation.compile.unsafe_build	boolean	true
cli.global.generation.evaluator.abolite_level	integer	0
cli.global.generation.evaluator.compile	boolean	false
cli.global.generation.evaluator.compile_serializations	integer or null	no serialized default
cli.global.generation.evaluator.disable_translation	boolean	true
cli.global.generation.evaluator.do_build	boolean	false
cli.global.generation.evaluator.do_build_replacement	boolean	false
cli.global.generation.evaluator.horizon_generations	integer	1
cli.global.generation.evaluator.iterative_orientation	enum	"up"
cli.global.generation.evaluator.max_integer_pair_cache	integer	1000000
cli.global.generation.evaluator.max_integer_pair_distance	integer	1000
cli.global.generation.evaluator.max_integer_scheme_variables	integer	500

Path	Type	Default
cli.global.generation.evaluator.n_integer	integer	1
cli.global.generation.evaluator.samplesExecutionMode	enum<ExecutionMode, Sequential", "MinResultRank"] ContractionMode>	
cli.global.generation.evaluator.stochastic	boolean	false
cli.global.generation.evaluator.submedian	boolean	false
cli.global.generation.evaluator.submedian_function_map	boolean	false
cli.global.generation.evaluator.tensorNetworkContractingOrder	enum<Order, "topdown", "bottomup", "lefttoright", "righttoleft", "arbitrary">	"arbitrary"
cli.global.generation.evaluator.verbose	boolean	false
cli.global.generation.feyngen.gamma_normalization	boolean	false
cli.global.generation.force_cuts	array<array<string>>	{}>
cli.global.generation.orientation	enum<Orientation, "left", "right", "top", "bottom", "none">	no serialized default
cli.global.generation.override_lmbooleans	boolean	false
cli.global.generation.threshold_submedian.assume_positive_external_energies	boolean	true
cli.global.generation.threshold_submedian.check_surface_at_generation	boolean	false
cli.global.generation.threshold_submedian.disable_integrated_ct	boolean	false
cli.global.generation.threshold_submedian.enable_thresholds	boolean	true
cli.global.generation.threshold_submedian.esurface_existence_threshold	number	1e-7
cli.global.generation.threshold_submedian.skip_thresholds_that_are_cuts	boolean	true
cli.global.generation.tropical_submedian.disable_tropical_generation	boolean	false
cli.global.generation.tropical_submedian.panic_on_fail	boolean	false
cli.global.generation.tropical_submedian.target_omega	number	1.0
cli.global.generation.uv.add_markers	boolean	false
cli.global.generation.uv.final_integral_dimension	enum<Dimension, "1D", "2D", "3D", "4D", "5D", "6D", "7D", "8D", "9D", "10D", "11D", "12D", "13D", "14D", "15D", "16D", "17D", "18D", "19D", "20D", "21D", "22D", "23D", "24D", "25D", "26D", "27D", "28D", "29D", "30D", "31D", "32D", "33D", "34D", "35D", "36D", "37D", "38D", "39D", "40D", "41D", "42D", "43D", "44D", "45D", "46D", "47D", "48D", "49D", "50D", "51D", "52D", "53D", "54D", "55D", "56D", "57D", "58D", "59D", "60D", "61D", "62D", "63D", "64D", "65D", "66D", "67D", "68D", "69D", "70D", "71D", "72D", "73D", "74D", "75D", "76D", "77D", "78D", "79D", "80D", "81D", "82D", "83D", "84D", "85D", "86D", "87D", "88D", "89D", "90D", "91D", "92D", "93D", "94D", "95D", "96D", "97D", "98D", "99D", "100D", "101D", "102D", "103D", "104D", "105D", "106D", "107D", "108D", "109D", "110D", "111D", "112D", "113D", "114D", "115D", "116D", "117D", "118D", "119D", "120D", "121D", "122D", "123D", "124D", "125D", "126D", "127D", "128D", "129D", "130D", "131D", "132D", "133D", "134D", "135D", "136D", "137D", "138D", "139D", "140D", "141D", "142D", "143D", "144D", "145D", "146D", "147D", "148D", "149D", "150D", "151D", "152D", "153D", "154D", "155D", "156D", "157D", "158D", "159D", "160D", "161D", "162D", "163D", "164D", "165D", "166D", "167D", "168D", "169D", "170D", "171D", "172D", "173D", "174D", "175D", "176D", "177D", "178D", "179D", "180D", "181D", "182D", "183D", "184D", "185D", "186D", "187D", "188D", "189D", "190D", "191D", "192D", "193D", "194D", "195D", "196D", "197D", "198D", "199D", "200D", "201D", "202D", "203D", "204D", "205D", "206D", "207D", "208D", "209D", "210D", "211D", "212D", "213D", "214D", "215D", "216D", "217D", "218D", "219D", "220D", "221D", "222D", "223D", "224D", "225D", "226D", "227D", "228D", "229D", "230D", "231D", "232D", "233D", "234D", "235D", "236D", "237D", "238D", "239D", "240D", "241D", "242D", "243D", "244D", "245D", "246D", "247D", "248D", "249D", "250D", "251D", "252D", "253D", "254D", "255D", "256D", "257D", "258D", "259D", "260D", "261D", "262D", "263D", "264D", "265D", "266D", "267D", "268D", "269D", "270D", "271D", "272D", "273D", "274D", "275D", "276D", "277D", "278D", "279D", "280D", "281D", "282D", "283D", "284D", "285D", "286D", "287D", "288D", "289D", "290D", "291D", "292D", "293D", "294D", "295D", "296D", "297D", "298D", "299D", "300D", "301D", "302D", "303D", "304D", "305D", "306D", "307D", "308D", "309D", "310D", "311D", "312D", "313D", "314D", "315D", "316D", "317D", "318D", "319D", "320D", "321D", "322D", "323D", "324D", "325D", "326D", "327D", "328D", "329D", "330D", "331D", "332D", "333D", "334D", "335D", "336D", "337D", "338D", "339D", "340D", "341D", "342D", "343D", "344D", "345D", "346D", "347D", "348D", "349D", "350D", "351D", "352D", "353D", "354D", "355D", "356D", "357D", "358D", "359D", "360D", "361D", "362D", "363D", "364D", "365D", "366D", "367D", "368D", "369D", "370D", "371D", "372D", "373D", "374D", "375D", "376D", "377D", "378D", "379D", "380D", "381D", "382D", "383D", "384D", "385D", "386D", "387D", "388D", "389D", "390D", "391D", "392D", "393D", "394D", "395D", "396D", "397D", "398D", "399D", "400D", "401D", "402D", "403D", "404D", "405D", "406D", "407D", "408D", "409D", "410D", "411D", "412D", "413D", "414D", "415D", "416D", "417D", "418D", "419D", "420D", "421D", "422D", "423D", "424D", "425D", "426D", "427D", "428D", "429D", "430D", "431D", "432D", "433D", "434D", "435D", "436D", "437D", "438D", "439D", "440D", "441D", "442D", "443D", "444D", "445D", "446D", "447D", "448D", "449D", "450D", "451D", "452D", "453D", "454D", "455D", "456D", "457D", "458D", "459D", "460D", "461D", "462D", "463D", "464D", "465D", "466D", "467D", "468D", "469D", "470D", "471D", "472D", "473D", "474D", "475D", "476D", "477D", "478D", "479D", "480D", "481D", "482D", "483D", "484D", "485D", "486D", "487D", "488D", "489D", "490D", "491D", "492D", "493D", "494D", "495D", "496D", "497D", "498D", "499D", "500D", "501D", "502D", "503D", "504D", "505D", "506D", "507D", "508D", "509D", "510D", "511D", "512D", "513D", "514D", "515D", "516D", "517D", "518D", "519D", "520D", "521D", "522D", "523D", "524D", "525D", "526D", "527D", "528D", "529D", "530D", "531D", "532D", "533D", "534D", "535D", "536D", "537D", "538D", "539D", "540D", "541D", "542D", "543D", "544D", "545D", "546D", "547D", "548D", "549D", "550D", "551D", "552D", "553D", "554D", "555D", "556D", "557D", "558D", "559D", "560D", "561D", "562D", "563D", "564D", "565D", "566D", "567D", "568D", "569D", "570D", "571D", "572D", "573D", "574D", "	

Path	Type	Default
cli.global.generation.uv.vakint.finfo	boolean	true
cli.global.generation.uv.vakint.fom	string	"form"
cli.global.generation.uv.vakint.mabool	boolean	true
cli.global.generation.uv.vakint.mabool	boolean	true
cli.global.generation.uv.vakint.mabool	boolean	true
cli.global.generation.uv.vakint.mabool	boolean	true
cli.global.generation.uv.vakint.mabool	boolean	true
cli.global.generation.uv.vakint.no	string	"MSbar"
cli.global.generation.uv.vakint.py	integer	1000000000000
cli.global.generation.uv.vakint.py	integer	10000
cli.global.generation.uv.vakint.py	boolean	true
cli.global.generation.uv.vakint.py	number	relative precision
cli.global.generation.uv.vakint.py	string	no serialized default
cli.global.generation.uv.vakint.py	string	"python3"
cli.global.generation.uv.vakint.run	integer	precision
cli.global.generation.uv.vakint.test	string	no serialized default
cli.global.generation.vector_polar	integer	"SingleGaugeAxial"
cli.global.log_style.full_line_sou	boolean	false
cli.global.log_style.include_fields	boolean	false
cli.global.log_style.log_format	LogFormat	"Long"
cli.global.log_style.short_timestamp	boolean	false
cli.global.logfile_directive	string	"off"
cli.global.n_cores.compile	integer	1
cli.global.n_cores.feyngen	integer	1
cli.global.n_cores.generate	integer	1
cli.global.n_cores.integrate	integer	1
cli.override_state	boolean	false
cli.state.folder	string	"./gammaLoop_state"
cli.state.name	string   null	""
cli.try_strings	boolean	true
runtime.general.additional_param_val	array<number>	[]
runtime.general.debug_cache	boolean	false
runtime.general.disable_flux_facto	boolean	false
runtime.general.enable_cache	boolean	false
runtime.general.evaluator_method	EvaluatorMethod	"SingleParametric"
runtime.general.generate_events	boolean	false
runtime.general.integral_unit	IntegralUnit	"auto"
runtime.general.load_compiled_cff	boolean	false
runtime.general.m_uv	number	1000.0
runtime.general.mu_r	number	1000.0
runtime.general.orientation_pat	path   string	no serialized default

Path	Type	Default
runtime.general.renormalization_localization_scale	number	1000.0
runtime.general.store_additional_weights_in_event	boolean	false
runtime.general.use_ltd	boolean	false
runtime.h_function.enabled_dampening	boolean	true
runtime.h_function.function	HFunction	"poly_exponential"
runtime.h_function.power	integer   null	no serialized default
runtime.h_function.sigma	number	1.0
runtime.integrand.h_function.enabled_dampening	boolean	no serialized default
runtime.integrand.h_function.function	HFunction	no serialized default
runtime.integrand.h_function.power	integer   null	no serialized default
runtime.integrand.h_function.sigma	number	no serialized default
runtime.integrand.n_3d_momenta	integer	no serialized default
runtime.integrator.bin_number_evolution	integer   null	no serialized default
runtime.integrator.continuous_dim_learning_rate	number	1.0
runtime.integrator.discrete_dim_learning_rate	number	1.0
runtime.integrator.integrated_phase	IntegratedPhase	"real"
runtime.integrator.max_prob_ratio	number	30.0
runtime.integrator.min_samples_for_integration	integer	1000
runtime.integrator.n_bins	integer	64
runtime.integrator.n_increase	integer	10000
runtime.integrator.n_max	integer	100000000000
runtime.integrator.n_start	integer	100000
runtime.integrator.observables_output_format	ObservableFileFormat	"json_mat">
runtime.integrator.seed	integer	69
runtime.integrator.show_max_wgt_info	boolean	true
runtime.integrator.target_absolute_number	number	no serialized default
runtime.integrator.target_relative_number	number	no serialized default
runtime.integrator.train_on_avg	boolean	false
runtime.kinematics.e_cm	number	64.0
runtime.kinematics.externals.data_label_by_index	array<integer   string>	[]
runtime.kinematics.externals.data_improvement_settings	ImprovementSettings	ImprovementSettings::Large_deformation_check
runtime.kinematics.externals.data_improvement_model	ImprovementModel	ImprovementModel::None
runtime.kinematics.externals.data_improvement_settings_only_warn_on_large_deformation	boolean	false
runtime.kinematics.externals.data_arrays	array<array<number>>	[]
runtime.kinematics.externals.type	string	"constant"
runtime.model	object	{}
runtime.observables	object	{}
runtime.quantities	object	{}
runtime.sampling.alpha	number	3.0



Path	Type	Default
runtime.sampling.b	number	1.0
runtime.sampling.coordinate_system	CoordinateSystem	"spherical"
runtime.sampling.graph_names	array<string>	[]
runtime.sampling.graphs	SumMode	"summed"
runtime.sampling.lmb_basis_ids	object	no serialized default
runtime.sampling.lmb_channel_weight	lmbChannelWeight	"ose"
runtime.sampling.lmb_channels	SumMode	"summed"
runtime.sampling.lmb_multichanneling	boolean	false
runtime.sampling.mapping	ParameterizationMapping	"Mapping"
runtime.sampling.orientations	SumMode	"summed"
runtime.sampling.power	number	1.0
runtime.selectors	object	{}
runtime.stability.check_on_norm	boolean	true
runtime.stability.escalate_if_exact	boolean	false
runtime.stability.levels	array<StabilityLevel>	{ "escalate_for_large_weight_threshold": 1e-7, "escalate_for_large_weight_threshold": 1e-7, "escalate_for_large_weight_threshold": 1e-7 }
runtime.stability.levels[].escalate_for_large_weight_threshold	number	no serialized default
runtime.stability.levels[].precision	precision	no serialized default
runtime.stability.levels[].required_precision_for_image	number	no serialized default
runtime.stability.levels[].required_precision_for_reconstruction	number	no serialized default
runtime.stability.loop_momenta_normalization_factor	number	1.0
runtime.stability.recording.recordable_stability_levels	boolean	no serialized default
runtime.stability.recording.recordable_momenta_escalation	boolean	no serialized default
runtime.stability.recording.recordable_results	boolean	no serialized default
runtime.stability.rotation_axis	array<object>	[{"alpha":0.1,"beta":0.2,"gamma":0.3,"type":"alpha"}]
runtime.stability.rotation_axis[].alpha	number	no serialized default
runtime.stability.rotation_axis[].beta	number	no serialized default
runtime.stability.rotation_axis[].gamma	number	no serialized default
runtime.stability.rotation_axis[].type	string	no serialized default
runtime.subtraction.disable_threshold	boolean	false
runtime.subtraction.esurface_existence_threshold	number	1e-7
runtime.subtraction.integrated_ct_settings.range.h_funct	boolean	function_settings.enabled_dampening
runtime.subtraction.integrated_ct_settings.range.h_funct_exp	string	"function_expansion"
runtime.subtraction.integrated_ct_settings.range.h_funct_sigma	integer   range	function_settings.sigma
runtime.subtraction.integrated_ct_settings.range.h_funct_sigma	boolean	function_settings.sigma
runtime.subtraction.integrated_ct_settings.range.type	string	"infinite"
runtime.subtraction.local_ct_settings.localisation	boolean	dynamic_width
runtime.subtraction.local_ct_settings.localisation	boolean	force_uv_dampers_to_one
runtime.subtraction.local_ct_settings.localisation	number	dynamic_width
runtime.subtraction.local_ct_settings.localisation	number	dynamic_width

Path	Type	Default
runtime.subtraction.overlap_settings.back_global_center	boolean	true
runtime.subtraction.overlap_settings.by_global_center	boolean	serialized default
runtime.subtraction.overlap_settings.clean_origin	boolean	true
runtime.subtraction.overlap_settings.clean_origin_all_files	boolean	false
runtime.subtraction.radial_root_resolution	number	64.0

## Version information

Save these identifiers with published results and include them in issue reports so that collaborators can reproduce the same software environment.

- **Documentation channel:** latest
- **Documentation route:** /products/gammaploop/latest/
- **Source revision:** 3edde1645d2cea206bce939b2d07ae0ba6766836

## Package versions and enabled features

- gammaploop/gammaplooprs — 0.3.4 (no optional features)
- gammaploop/gammaploop-api — 0.1.0 (features: cli)
- gammaploop/gammaploop — 0.3.4 (features: python\_api, python\_abi, python\_stubgen)
- linnet/linnet — 0.17.0 (features: nodestore-vec, serde, bincode, drawing, symbolica)
- linnet/linnet-py — 0.1.0 (features: extension-module, abi3-py310)
- spenso/spenso — 0.6.0 (features: shadowing)
- spenso/spenso-macros — 0.3.2 (features: shadowing)
- spenso/spenso-hep-lib — 0.1.7 (no optional features)
- spenso/spynso3 — 0.1.2 (features: python\_stubgen)
- idenso/idenso — 0.3.0 (features: bincode, reference-cases)
- idenso/idenso — 0.3.0 (features: python, python\_stubgen)
- vakint/vakint — 0.1.2 (no optional features)
- vakint/vakint — 0.1.2 (features: symbolica\_community\_module, python\_stubgen)